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April 1st, 2025

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Dear Dr. Abdullah and Ms. Saha,

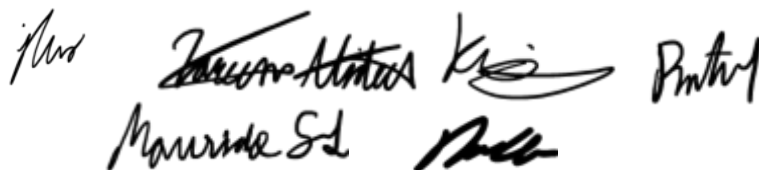
Our team at AMBUTECH have prepared and submitted our final report titled *Power-Assisted Gurney*. This report was prepared to present a finalized solution aimed at preventing musculoskeletal injuries endured by paramedics from the repeated pushing and pulling of patients using conventional gurneys. The Power Assisted Gurney was chosen through a comprehensive evaluation involving a weighted decision matrix, evaluation scales, and sensitivity analysis.

As your organization oversees the Board of Directors responsible for each provincial and territorial access to standard equipment and technology, our proposed solution aligns with the Canadian Standard for Paramedic Ground Emergency Response Vehicles and Equipment to ensure compliance, safety, and practical integration. This innovation is intended to improve the longevity and well-being of frontline emergency responders, enabling them to serve more effectively and sustainably.

This report outlines the identified problem, presents conceptual designs and their evaluations, and provides detailed documentation of the mechanical, electrical, and software components, as well as an updated work plan and resource considerations.

For any questions regarding the administrative nature or technical contents of this report, please connect with the team. Your interest and support are greatly appreciated.

Sincerely,
AMBUTECH Team



POWER-ASSISTED GURNEY



ENGG*3100 W25

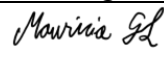
Group 26

April 1st, 2025

University of Guelph

Signature Page

In signing this report, I certify that I have been an active member of the team and provided approximately equal contribution to the work. I take shared credit and responsibility for the content of this report. I understand that taking credit for work that is not my own is a form of academic misconduct and will be treated as such.

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* Include where/how AI tools were us

EXECUTIVE SUMMARY

AMBUTECH has focused on developing a solution to reduce physical toll experienced by paramedics when transporting patients on standard gurneys. Repetitive pushing and pulling motions during emergency calls place workers at risk of musculoskeletal disorders (MSDs), contributing to early career-ending injuries. This report presents the outcome of a design process aimed at addressing that risk through the development of a powered gurney system that eases physical strain without compromising safety or functionality.

Early stages of the project involved identifying user needs and technical constraints through research and stakeholder mapping. From there, four conceptual solutions were generated and compared using a weighted decision matrix and sensitivity analysis. Based on key criteria—maneuverability, ease of use, durability, battery performance, and cost—the Power-Assisted Gurney was selected for final development.

The design integrates motorized wheels, throttle-based speed control, and a 24V lithium-ion battery system mounted to an aluminum frame. Safety features such as a manual override and emergency shutoff ensure that the system remains reliable in unpredictable environments. The full assembly is built to comply with Canadian Standards for Paramedic Ground Emergency Response Vehicles and Equipment.

A detailed bill of materials including the frame, motors, sensors, battery pack, control hardware, and enclosure estimates the cost to build a single unit at \$1,570.22 CAD. When scaled to 150 units over one year, the total implementation cost which includes materials, labor, equipment, and overhead is projected to be \$958,922 CAD.

A comprehensive lifecycle analysis was conducted using a cradle-to-grave approach. Total emissions from material production were calculated at 1,172.84 kg CO₂, while transportation of components contributed an additional 191.54 kg CO₂. Over its expected operational lifespan, emissions from system use were estimated at 3,390.86 kg CO₂, primarily from the DC motors and battery pack. These figures highlight operations as the most emission-intensive stage, reinforcing the value of future efficiency improvements and renewable energy integration.

This report documents the complete development process, final system design, associated costs, environmental considerations, and justification for the selected concept. The Power-Assisted Gurney offers a meaningful step forward in reducing workplace injuries in emergency medical services while remaining economically and environmentally responsible.

ACKNOWLEDGEMENTS

For their valuable guidance and feedback throughout the development of this project, it is our pleasure to express our gratitude to Dr. Hussein Abdullah and Puja Saha. Furthermore, the research and insights provided by real-world paramedics and external EMS workers who participated in relevant ergonomic studies through their feedback and efforts. Lastly, we acknowledge the contributions of the AMBUTECH team, whose efforts and collaboration made it possible to design and develop this project to assist those in need within ambulance services.

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LIST OF SYMBOLS AND ACRONYMS

SYMBOLS AND ACRONYMS		EXPANSION
EMS		Emergency Medical Services
MSD		Musculoskeletal Disorders
SA		Sensitivity Analysis
OPPAT		Ottawa Paramedic Physical Abilities Test
OBC		On-Board Computer
DDC		Dynamic Damper Control
AGV		Autonomous Guided Vehicle
SLAM		Simultaneous Localization And Mapping

INTRODUCTION

PROBLEM DESCRIPTION

The lack of ergonomic support in current emergency medical equipment design fails to accommodate the fast and high-intensity movements required by paramedics, increasing the risk of musculoskeletal injuries and, over time, contributing to chronic health issues that may reduce workforce efficiency and compromise patient care. Poorly designed equipment is a widespread issue in the healthcare industry. Emergency Medical Services (EMS) personnel, particularly paramedics, are heavily affected, as their tight time constraints and high-stakes responsibilities prevent them from using proper movements during their routine tasks. An effective ergonomic environment aims to reduce or remove potential sources of injury or discomfort while maintaining an efficient workflow [1]. Paramedics engage in a variety of tasks such as pushing/pulling gurneys to transport patients to an ambulance or hospital, carrying heavy equipment, and other physically demanding activities. Over time, these physical activities can lead to negative health effects in the long term if equipment isn't designed to support needs.

Excessive movements and strain can result in fatigue, injuries, and musculoskeletal disorders (MSD), all of which can negatively impact paramedic performance and contribute to high turnover rates. MSDs are painful injuries or disorders affecting muscles, joints, and other parts of the body that are prevalent among EMS workers [2]. Over time, these conditions can compromise the healthcare system by jeopardizing both the quality of patient care and overall safety.

Paramedics operate in a high-stakes environment and require an intuitive solution that minimizes injury risk during patient transfer while integrating seamlessly into the healthcare system. Patient transfers are both frequent and physically demanding, with the repetitive pushing and pulling of gurneys identified as one of the main causes of health problems among paramedics. Reducing the strain and injuries related to pushing and pulling gurneys will be the focus when designing a solution. The Why-Why diagram in Appendix B outlines EMS worker tasks and the potential side effects. Patients must be quickly transported to an ambulance or hospital to receive the necessary care, without compromising the health of paramedics. To maximize impact, the solution needs to address the physical demand of pushing or pulling patients during emergency situations. An ergonomically designed solution to pushing and pulling strains would reduce physical injuries for paramedics when relocating patients in emergency scenarios.

An effective ergonomic solution such as the redesign of EMS equipment would address the social, environmental, and economic aspects that lead to negative health effects in paramedics when pushing or pulling gurneys.

The relevant stakeholders for implementing the solution are outlined in Appendix A. Paramedics will have access to a new tool that will help them transport patients, reduce strain on their hands and body, and reduce the risk of injury. Failing to implement a solution to support paramedics will result in high injury rates and strain on emergency services due to staff shortages. The next stakeholders are government agencies and hospital administration. Both organizations will be positively affected as improved equipment will lead to efficient operations, patient satisfaction, and a more resilient workforce due to reduced injury rates. A potential downside for these organizations is the cost involved in implementing the solution; however, the long-term benefit of improved equipment will lead to lower medical expenses from workplace injuries. Using existing equipment will lead to increased spending on medical coverage, training, and rehabilitation of injured paramedics. For the public and patients, better-equipped paramedics will improve the quality of service and response times from emergency staff. Not implementing better equipment could potentially compromise the lives of the public during emergency situations if paramedics are physically limited or injured. The last group of stakeholders are the ergonomic consultants and Union Representatives; both groups would support and benefit from improved equipment. Fewer injuries in the working sector will lead to a better quality of life for workers and improved work satisfaction.

The solution can be implemented with existing materials and manufacturing processes for medical equipment, leading to a smooth integration into emergency services. The main environmental impact of medical equipment stems from the manufacturing and use of medical equipment, the production and use of medical devices contribute to about 4.4% of global carbon emissions [3]. These emissions do not only damage the environment but can also indirectly harm public health. The materials involved in the manufacturing of existing gurneys involve metals and plastics. Aluminum and stainless steel are the best choices as these provide a good balance between strength and weight [4]. Additionally, the durability of these materials makes them excellent choices for emergency scenarios to ensure the equipment is reliable. The popularity of these materials in various industries including the medical sector, makes them easily accessible for manufacturers. Unfortunately, the extraction of both aluminium and steel often involves deforestation, wildlife loss, and severe health impacts on workers of mining companies [5]. Although the environmental impacts of medical equipment have a significant effect on the environment, the benefit of improving emergency services and the health of workers greatly outweighs the negatives.

Reducing workplace injuries leads to various cost-reducing benefits for Hospitals. As previously mentioned, improved equipment translates to lower medical and compensation expenses, improved worker performance, and lower turnover rates. The price of a typical ambulance gurney can range from \$6,000 to \$15,000 CAD [6] while high-tech and higher quality gurneys can go up to \$27,000 CAD [7]. The ideal price for the new product should be lower or near the cost of a typical ambulance gurney. This price range should ensure both large and small health organizations have access to the new product.

BACKGROUND INFORMATION AND LITERATURE REVIEW

In emergency situations, paramedics frequently engage in the act of pushing or pulling ambulance gurneys or stretchers to transport immobilized patients suffering from life-threatening injuries into ambulances and hospitals. It is a common responsibility for paramedics to ensure that patients are transported quickly and safely. However, this constant, repetitive action of pushing and pulling poses an ergonomic risk for paramedics.

Rigorously pushing and pulling of gurneys and stretchers repeatedly, especially with heavier patients, results in serious injuries causing a long-term effect on overall health and performance of paramedics. These actions, when endured on a daily basis, impose a high ergonomic risk that can negatively impact the musculoskeletal system due to increased stress or force [8]. For pushing and pulling activities, it was reported that the lower back and shoulders are the most common pain areas. A study from the Netherlands found that individuals whose jobs involved frequently pushing or pulling loads greater than 50kg (100lbs) were roughly twice as likely to experience back pain than individuals without such job requirements [9]. Additional studies involving seven different services across Canada found that paramedics identified stretcher loading and unloading (25.6% of respondents), carrying equipment (19.5%), and pushing and pulling the stretcher (13.4%) as the most physically demanding tasks [10].

There are several existing solutions that have been developed to assist paramedics in transporting patients on gurneys and stretchers to prevent injuries and focus on improving ergonomics. The emergence of powered stretchers, like Stryker's Power-Pro XT and Ferno's Power X1 Ambulance Cot, has proven to be beneficial for EMS workers. Its innovative battery allows the lifting and lowering of the cot using a button for easy and efficient loading. They also feature a 700lb unassisted lift capacity, which can reduce the strain of bariatric transport [11]. However, there are limitations present among the assistive benefits of powered stretchers. Factors such as increased weight, higher costs, maintenance, and repairs are a few that could potentially be a challenge and difficult to access for smaller,

underfunded emergency services. Additionally, these powered stretchers only assist with lifting and loading, not the pushing and pulling of equipment over a distance. Ergonomic Pull Handles are among those that do assist in pushing and pulling medical equipment. The Rowland Ergonomic Cot Pull Handle is a device specifically designed for Stryker's Power-Pro XT. It promotes proper posture and pulling techniques, resulting in a minimized risk of back strain and other injuries associated with manual stretcher transport. The handle offers better control of the stretcher's movement to ensure the safety of the patient as well as comfort during transport. Although this can help with pulling, it still requires manual pushing/pulling to transport the equipment and patient. Furthermore, it is limited to certain terrains as it doesn't significantly help on rough surfaces like gravel, grass, or snow. Lastly, there are already existing solutions involving motorized stretchers for assisting in patient transportation. Stryker's 1125 Zoom Prime Series Motorized Stretchers are designed to reduce physical strain on clinicians, enhance patient comfort, and improve hospital efficiency. The stretcher's functionality gives patients the independence to adjust themselves without calling for caregiver assistance. Its motorized drive system generally reduces strenuous pushing and pulling. Despite their benefits, these stretchers are only primarily used in hospitals. They are not designed for EMS use as they are not optimized for outdoor or rough terrain. Difficulty moving in tight spaces due to their weight and bulkiness, as well as high initial costs, makes these stretchers not the ideal solution to transport patients to an ambulance on the field.

To develop a feasible solution to be used for paramedics in the field, it must comply with the necessary standards, codes, and guidelines. These standards were important to identify as they would act as parameters when developing a design in the design process. These parameters were used to comply with ergonomic guidelines, paramedic equipment design standards, and federal/provincial safety codes. In general work situations, there are limits that should not be exceeded to prevent injuries, including those stated previously. Pushing and pulling tasks are safer if they require smaller forces, especially if the task requires the hands to be above the shoulder or below the waist level, exerting a force for longer than 5 seconds, or exerting a force at an angle not directly in front of the body [12]. Appendix C displays the *Recommended Upper Force Limits for Horizontal Pushing and Pulling*. In addition, the *AORN Ergonomic Tool* document was utilized to observe the recommended force for pushing stretchers [13]. Considering these guidelines ensures our solution is proven to be ergonomic. Furthermore, the Ontario Ministry of Health provided the Provincial Equipment Standards for Ontario Ambulance Services, a document which compiles information on the standards and codes for every medical equipment used [14]. The development of the structural design will be tailored to meet these equipment standards, including stretcher restraints, lift assist, adjustable breakaway, minimum load-

bearing requirement. etc., so that it may be legally used for EMS workers in the field. The list of relevant design standards from the document can be found in Appendix D.

SCOPE OF PROJECT

The objective of this project is to develop a theoretical design solution aimed at reducing the physical strain and risk of injury faced by paramedics during patient transport. The power-assisted gurney was selected as the most viable option through detailed analysis, including ergonomic assessment, design evaluation, and theoretical modelling.

Having completed the first three phases of the NASA JPL Design Process—Understand, Explore, and Define [15], the team now looks ahead to the next critical steps: prototyping, real-world testing, and refinement. The proposed Power-Assisted Gurney integrates motors, load sensors, and microcontrollers to reduce the effort required for pushing and pulling, especially on inclines and rough terrain. The next phase will involve building a working prototype, conducting battery and durability tests, and performing simulations under emergency conditions to validate its performance.

As part of future development, AMBUTECH plans to partner with EMS providers and hospitals to pilot the gurney in controlled environments and collect end-user feedback. This will inform iterative improvements to usability, durability, and training design [16]. Regulatory considerations, manufacturing feasibility, and cost optimization will also be addressed to support potential commercialization.

This scope does not include changes to existing EMS protocols or general paramedic training programs. Broader factors such as mental health, workplace stress, or unrelated injury causes are considered outside the scope. By narrowing the focus to assistive transport tools, the project remains targeted and actionable, with a long-term vision of contributing to safer, more efficient paramedic operations.

The ongoing goal is to transition this design from theory to field-ready solution, providing value to both EMS professionals and patients through thoughtful innovation and engineering.

CONSTRAINTS AND CRITERIA

When designing assistive technology to improve ergonomics for EMS First Responders, there are various key constraints and criteria, listed in Table 1 and Table 2 respectively. These constraints and criteria will guide the team's design solution.

Table 1: Design Constraints

Constraints	Justification
Supporting a Minimum Weight	The device must support a minimum weight of 250lbs. This ensures accommodation of a wide range of patient body types, aligning with ergonomic standards for medical equipment [17]
Standard Device Sizing	The device must fit within standard ambulance storage spaces, not exceeding 72 inches in length and 24 inches in width. Ensuring compatibility with existing ambulance configurations ensures seamless integration and accessibility during emergencies [18].
Long Term Device Functionality	The device must withstand daily use in various environmental conditions, maintaining functionality over a minimum lifespan of 5 years. Long-term reliability is essential for cost-effectiveness and consistent performance in diverse emergency settings [19].
Adhering to Standards	The design must adhere to relevant standards, such as the Canadian Standard for Paramedic Ground Emergency Response Vehicles and Equipment (CSA D500:20). Compliance ensures the device meets established safety and performance benchmarks [18].
Ergonomic Efficiency	The device must allow operation without awkward postures, excessive reaching, or repetitive motions. Adhering to ergonomic principles minimizes the risk of MSDs among paramedics [17].
Safety Mechanisms	The device must incorporate safety mechanisms such as locking systems and stability controls to prevent accidental injuries. Protects both patients and

	paramedics, ensuring secure operation during pushing and pulling tasks [18].
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Table 2: Design Criteria

Criteria	Justification
Minimize cost	The device should be as cost-effective as possible without compromising safety and durability. Ensuring affordability for EMS departments while maintaining essential functionality.
Minimize training requirements	The device should have an intuitive design that requires minimal training. This reduces onboarding time and ensures quick adoption by EMS personnel.
Minimize weight	The device should be as lightweight as possible while maintaining durability. A lightweight design reduces the physical burden on EMS personnel, promoting ease of use in various emergency scenarios [19].
Minimize cleaning time	The device should be constructed from materials that are easy to clean and disinfect. Ensuring compliance with infection control protocols and reduces downtime due to maintenance.
Maximize battery efficiency	The device should operate for a full EMS gurney deployment without requiring a charge. Preventing workflow disruption and enhances reliability during long deployments.

The team has also made some assumptions as seen below in Table 3.

Table 3: Assumptions

Assumption	Justification
Standardized Ambulance interior layouts.	Assuming uniformity in ambulance design allows for a design that fits seamlessly into existing spaces.
Uniform EMS personnel work shifts.	Designing for 12 hour extended use periods ensures the device remains functional and reliable throughout typical shifts.

Urban and Rural device usage	Considering diverse operational environments ensures the device’s versatility and effectiveness in various scenarios.
Temperature thresholds	Designing for a temperature range of -20°C to 40°C ensures functionality in diverse Canadian climates.

DESIGN PROCESS AND SOLUTION

PRELIMINARY INFORMATION

Our team required an understanding of relevant standards and codes before moving forward with our design process. These standards and codes exist to protect patient and paramedic safety, which falls within our scope.

Provided by the Ontario Ministry of Health, the Provincial Equipment Standards for Ontario Ambulance Services [14]. offers a compilation of information on the standards and codes for medical equipment, including Cots and Stretchers/Accessories. Appendix C lists the provincial equipment standards relevant to our design process.

Our designs will also take inspiration from existing stretchers and cots. We intend to create a solution that is operated similarly to current cots and stretchers, to minimize the operator’s learning curve. Relevant information includes manuals of previous existing solutions of powered cots and gurneys such as Stryker’s Power Pro Cot [20].

In addition, paramedics have physical requirement guidelines that follow the Ottawa Paramedic Physical Abilities Test (OPPAT), an occupational test for frontline paramedic [21]. This material provides a baseline for what is expected from paramedics, which will be used to inform our assumptions throughout the design process.

TOOLS USED

The tools used for this project are described in Tables 4 and 5.

Table 4: Description of Tools

Design Stage	Tool	Justification
Problem Identification	Why-Why Diagram	Our why-why diagram (Appendix A) helped us define the root cause of our problem.

Solution Brainstorming	Research Materials – Journals, Articles, Papers	Research materials were collected to inform the problem definition, constraints and criteria, potential solutions and throughout the design defense.
Model/Prototype Construction - Mechanical	Solidworks, GrabCAD, McMasterCarr	We used SolidWorks for all 3D-modelling related work (creating prototypes and models for simulation). This was the best choice because most of the team has SolidWorks experience. GrabCAD and McMasterCarr offer a variety of free 3D models that we used for standard components such as motors and gearboxes. We used Solidworks Simulation to run FEA simulations for structural elements.
Model/Prototype Construction - Calculations	MATLAB	Matlab was used to run torque calculations and generate plots. All other calculations were done by hand and reviewed by the team.
Model/Prototype Construction - Electrical	KiCAD	KiCAD is a tool used to simulate and design electric circuits. We used this to design our PCBs to minimize their form factor.
Model/Prototype Construction - Costing	McMasterCarr	Apart from helping with free 3D models, McMasterCarr helped us generate an estimate of the material costs to produce a prototype.

Table 5: Description of Tools Continued

Design Stage	Tool	Justification
Project Management	Microsoft Teams, LettuceMeet	- Microsoft Teams was our primary file-sharing and storage tool. We used teams to organize the project into a set of 5 deliverables. Relevant design files such as CAD, simulation results, referenced images and research documents were compiled to ensure accessibility by the rest of the team. - LettuceMeet was used to monitor team member schedules for the purpose of finding optimal times to schedule meetings. - Finally, in-person team meetings served to quickly get the entire team up to speed and discover issues in

		project progress quickly, so that they could be addressed. The team met anywhere from 1 to 3 times a week to discuss reports, share progress or work on presentations.
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THE DESIGN PROCESS

Our team decided to use the NASA Jet Propulsion Laboratory (JPL) design process. The first phases of this process involve problem identification, solution brainstorming, and design selection [15]. The iterative phase includes a cycle of prototyping, testing, optimization and sharing. A process diagram is visible in Figure 1 below.

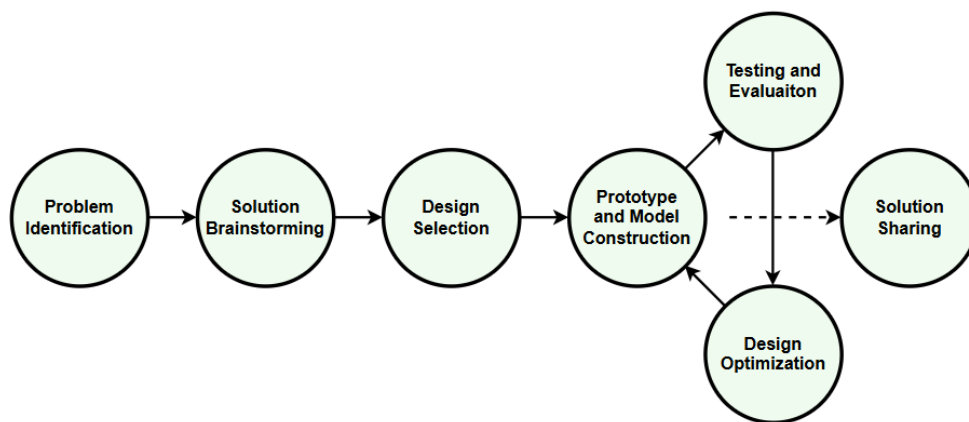


Figure 1: NASA JPL Design Process

The following sections are a review of our team’s experience in each step of the NASA JPL design process. Our brainstorming techniques are described, potential solutions are listed, and our selection methods are explored. Our selected design is explained in detail, followed by a discussion of our revisions thus far.

BRAINSTORMING

Having identified our problem as described in Section 1, our team moved on to brainstorming. Various brainstorming techniques were applied to generate potential design solutions, mainly the SCAMPER method and the “Day in the Life” strategy, also known as the Shift Perspective method.

The SCAMPER method [22] stands for substitute, combine, adapt, modify, put to another use, eliminate, and reverse. This method focuses on shifting perspectives from a head-on problem-solving method to a creative one. It allows us to look at the existing solutions and technology and ask if it could be improved/applied by being simplified, used in new ways, combined, or by making part/material/process changes.

The “A Day in the Life” strategy involves stepping into the shoes of stakeholders and imagining their daily experiences in detail [23]. Understanding the daily challenges of the stakeholders, such as paramedics or hospital management staff, involves observing them during their daily responsibilities, conducting interviews or surveys, and listening to their personal narration of events. This approach integrates customer-centric thinking into the design process, allowing the design to be tailored with the customer’s work, operations and satisfaction in mind.

For our problem, the main stakeholders involved are paramedics, hospital management, and government agencies. Each of these groups have different factors that shape their viewpoints; Appendix E contains a Shift Perspective Analysis Diagram that gives a high-level view of the perspective of each stakeholder.

The perspective of paramedics, our most important stakeholder, is greatly influenced by job satisfaction and ergonomic requirements. Although the role of a Paramedic is stressful, with some of the highest rates of injuries and illnesses of all occupations [24], many members find the societal benefit fulfilling - EMS provides a positive impact in communities during crisis. To continue providing for the community, workers need access to reliable equipment for daily operations. Therefore, our solution must be reliable.

Our solution should also minimize the physical strain on paramedics. Ergonomic design, assistive technologies and training are methods of reducing EMS staff injury and strain. Ergonomic design refers to designing equipment with the goal of reducing or removing a possible hazard, while assistive technology refers to equipment that enhances operations and make tasks easier to perform.

In the case of hospital management staff, their perspective is shaped by their interest in lowering costs, raising staff productivity, and preserving the hospital’s reputation. They want solutions that provide high-quality equipment at low prices, while also minimizing maintenance. Staff productivity must be prioritized, so equipment should also be user-friendly, reliable, and integrated into the hospital’s workflow to minimize training time and disruptions. To preserve the hospital’s reputation, only reliable and safe equipment should be used. To ensure our design meets these needs, we must focus on cost-effective manufacturing and maintenance, ergonomic designs to improve staff productivity, and ensure the design follows regulations and standards of government agencies to stay credible and trustworthy.

Government agencies, such as Health Canada and The Ontario Ministry of Health, have a perspective shaped by public safety, economic savings, and policy leadership. Their primary focus is ensuring that medical ambulance equipment meets strict safety and performance standards to protect both patients and healthcare providers. These

organizations require that all devices follow a set of standards such as The Provincial Equipment Standards for Ontario Ambulance Services[14] and regulations for Medical Devices by the government of Canada [25]. This ensures that emergency medical equipment functions properly during life-or-death scenarios, and risks are minimized.

Government agencies also aim to reduce costs in healthcare while maximizing value. This means ambulance equipment should be durable, easy to maintain, and have a long service life to avoid frequent replacements. Ergonomic equipment also means workers will be less likely to develop health conditions, lowering medical compensation costs. Additionally, agencies favor designs that allow for efficient resource allocation that can be used across multiple emergency services. The solution must adhere to government standards and regulations, and it must be cost effective.

Government organizations want to establish clear rules and industry norms that encourage creativity while upholding responsibility. By pushing manufacturers to provide more effective, eco-friendly, and cutting-edge ambulance equipment, they hope to take the lead in healthcare innovations. To meet these demands, the design should prioritize domestic and international safety regulations, economical manufacturing without compromising on quality, and the incorporation of contemporary technology like smart monitoring systems or telemedicine capabilities.

ALTERNATIVE SOLUTIONS

Conceptual Design 1: Smart Suspension Gurney

The Smart Suspension Gurney, seen in Appendix F, uses electronically controlled dampers as the wheel suspension system, allowing for real-time terrain adaptation for improved stability, maneuverability, and ergonomic handling. Sensors detect patient weight, rough surfaces, and uneven terrain, sending the data to an on-board computer that dynamically controls the suspension to minimize vibrations. Reduced vibrations will decrease impact forces and paramedic fatigue, making transport smoother. The design is functional for both indoor and outdoor use, however, the system requires an OBC, battery, and four damper units, which increases the weight. It will also have to fit within dimension tolerances and have a minimum battery life. The Shift Perspective method was used to consider both patients and paramedics, reducing vibration for patient comfort and easing the physical strain on responders. This dual-focus approach improves usability for both stakeholders. Appendix F how the design meets constraints and criteria.

Conceptual Design 2: Autonomous Guided Vehicle-Assisted Gurney

The Autonomous Guided Vehicle Assisted Gurney, shown in Appendix F, would be a combination of existing AGV technology, typically used in the manufacturing/materials handling industries, with patient hospital beds and gurneys for assisted or autonomous pushing and pulling. Similar technology already exists for materials handling at Toronto's Humber River Hospital [26]. This idea was generated using the SCAMPER method, specifically C – Combine. AGVs use sensors [27] to follow path lines using magnetic tape, painted lines, or inductive wire. They also have a variety of safety sensors on board, including force sensors for collision detection, ultrasonic sensors for distance, and LiDAR for navigation and obstacle detection. Simultaneous localization and mapping technology is also available to map environments, plan paths and detect obstacles [28]. An AGV would be requested by a nurse or doctor and would route itself from a charging station into the requesting room. It would then attach itself to the underside of the gurney and begin moving to the required location. The AGV could provide most of the force required to move the gurney at a safe speed while being supervised by a nurse/doctor during travel. Appendix F discusses how the design meets constraints and criteria.

Conceptual Design 3: Adapting-Handle Gurney

The Shape-Adapting Handle Gurney is designed to improve paramedic ergonomics while ensuring patient comfort. This gurney features a grip handle that molds to the user's hand, reducing wrist strain and fatigue during transport. After each use, the handle returns to its original form, ensuring adaptability for multiple users. Additionally, smart cushioning is added to the gurney's surface to provide padding to enhance patient comfort. A concept sketch for this design can be found in Appendix F. The ergonomic handle is made from a flexible, memory-forming material, allowing for a customized grip that improves control and minimizes hand fatigue. This reduces injuries and the likelihood of MSDs among paramedics while ensuring safe and efficient patient transport. Meanwhile, smart cushioning technology is designed to alleviate pressure points, improving the overall patient experience by reducing discomfort over extended durations. The gurney's frame is built to meet government durability standards, ensuring maneuverability without adding excessive weight. By improving ergonomics, this gurney will reduce EMS staff hand fatigue and wrist strain, leading to better handling and improved health. The integration of smart cushioning also ensures a more comfortable experience for patients, while its lightweight design keeps it comparable to existing gurneys, preventing extra strain on healthcare workers. However, the use of advanced materials may lead to higher production costs, making it more expensive to manufacture. Additionally, the durability of the shape-adapting handle

requires continuous reliability testing to ensure it can withstand frequent use in an emergency environment. The smart cushioning may also wear down over time, which will require occasional maintenance or replacement. The Shift Perspective method played a key role in shaping this design by examining the gurney from both the Paramedic's and the patient's perspectives. Traditional gurneys focus mainly on patient safety and comfort, but by shifting our perspective to include staff needs, there is a way to improve ergonomics and ease of use. This approach led to the development of a solution that reduces MSDs among Paramedics while maintaining a high standard of patient comfort. Appendix F below explain how the design meets constraints and criteria.

Conceptual Design 4: Power-Assisted Gurney

The Power-Assisted Gurney is a motorized gurney that helps paramedics with patient handling and transportation to and from an ambulance. With its motorized features and data-collecting sensors, it can relieve and reduce strain on a paramedic's body and can be used in various scenarios transporting different cases of patients in different terrains. The Power-Assisted Gurney consists of motorized wheels help either to push or pull the gurney. The wheels are powered based on patient weight and operator control and can easily transition from push to pull or vice-versa. A weight sensor pad is used to detect and collect the patient's weight, and a throttle is used to allow paramedics to control the power of the wheels. Powered wheels are located either at the front or back of the gurney. The computer, located underneath the gurney, collects data (input) from the weight sensor pad and throttle. It then outputs a safe maximum power to the wheels. To power the computer, sensors, and other important features, a battery pack is used. It can be rechargeable for re-utilization. In terms of safety, the Power-Assisted Gurney will return to manual mode when the battery runs out. In addition, power is not provided to the wheels until the throttle is used. A gurney consisting of a computer, sensors, and motorized wheels can be costly compared to a normal gurney. This limits its accessibility, especially for smaller, lower-funded medical institutions. Maintenance is another concern due to its complex mechanisms and features. Given the consideration that these gurneys will be used extensively and aggressively in pressurized scenarios and terrain, a consistent check-up of this device is mandatory to ensure the gurney can assist the paramedics, but ultimately safely transport the patient. Lastly, the added components will increase the gurney weight compared to that of a normal gurney. This increased weight would require paramedics more effort, especially in manual mode. The Shift Perspective was used to generate this idea. This method focused on the perspective of the paramedics as they are the primary operators of the power-assisted gurney. With this perspective, this alternative solution emphasized on reducing the strain on

paramedics by creating a design that improves ergonomics. To further reduce strain, this solution includes assistive technology that would be able to support paramedics when handling the gurney. These considerations with the use of shifting perspectives led to a power-assisted and motorized solution with control. Appendix F below explain how the design meets constraints and criteria.

DESIGN SELECTION

Design selection incorporates the use of a weighted decision matrix and sensitivity analysis to evaluate each design alternative. With this approach, it will be easier to assess, evaluate, and ultimately select the preferred design.

Weighted Decision Matrix

A weighted decision matrix is a quantitative technique that compares and assesses multiple options against a set of criteria weighted by level of importance [29]. This is useful when there are multiple decision criteria to consider, of varying levels of importance. In our case, we will use a weighted decision matrix to select the most suitable design.

To create and utilize a weighted decision matrix, we must first list our options, as we have done in the previous section. Once options are listed, we must select our criteria – we will be using 5: maneuverability, cost, durability and maintenance, ease of use and training, and battery life. The criteria were derived from previous sections and refined for the sake of our decision matrix.

One of the most important steps for creating a weighted decision matrix is assigning weightings to each of our criteria [30]. Naturally, every weighting must be justified. Appendix G discusses the justification for each criterion assigned weighting.

After determining our criteria and assigning justified weightings, we move towards rating each design option per each criterion. To produce the most objective result, a defined evaluation scale must be created and clearly described. The evaluation scale will standardize criterion ratings and define what a poor, good or excellent score represents. Appendix H contains the evaluation scale for each criterion.

With the evaluation scale complete, the design solutions can now be rated for each criterion, and the final weighted scores can be compared. The final weighted score for each option is listed in Appendix I.

The decision matrix led the team to believe that the Power Assisted Gurney was the most suitable design. However, to make sure this was the case, a sensitivity analysis was conducted to measure the sensitivity of our results to variations in the criterion weightings.

Sensitivity Analysis

The 2nd phase of our design selection process is a sensitivity analysis. Our goal was to identify the sensitivity of our results with respect to our chosen criteria weightings [31]. To do this, we modified our weightings into 4 unique weighted decision matrices. The ratings remain identical, but the final scores change according to the adjusted weightings. A summary of the procedure is detailed in Appendix J. Appendix K demonstrates the weightings for all 5 decision matrices.

The team decided to perform a standard deviation analysis (equation 2.1 below) for each design option. A standard deviation calculation is performed to show which solution has the least amount of variance in scores across all the sensitivity analyses. The findings of the sensitivity analysis and the standard deviation calculation and detailed sensitivity analysis tables are available in Appendix J.

$$\sigma^2 = \frac{\sum_0^i (x_i - \bar{x})}{n - 1}$$

The findings clearly showed that the Power Assisted Gurney was the best option overall and exhibited the least variation (was the least sensitive) when weightings were adjusted, therefore making it resistant to subjective criteria weightings. It is the best choice overall, irrespective of which criteria are prioritized.

DETAILED FINAL DESIGN

Overall Design

The Power-Assisted Gurney, seen in Figure 2, aims to assist operators, such as paramedics, improve their ergonomics and reducing the risk of bodily strain of pushing and pulling through its various supportive features. Compared to other existing powered stretchers used in ambulance services, such as Stryker's Power-Pro XT and Ferno's Power X1 Ambulance Cot, this product utilizes data-collecting sensors, a microcontroller, hand-held controls, motorized wheels, and a battery source to power the system to ensure

paramedics can transport patients with this gurney through the action of pushing and pulling. This system incorporates sensors to detect the weight of the patient and adjust the motor assistance accordingly to ensure smooth and efficient operation. Intuitive controls will allow the paramedics to regulate the power and regulate the level of assistance while maintaining full control over movement. Safety features, including emergency stops and zero-point turning, are integrated into the design to increase user confidence and avoid accidents.

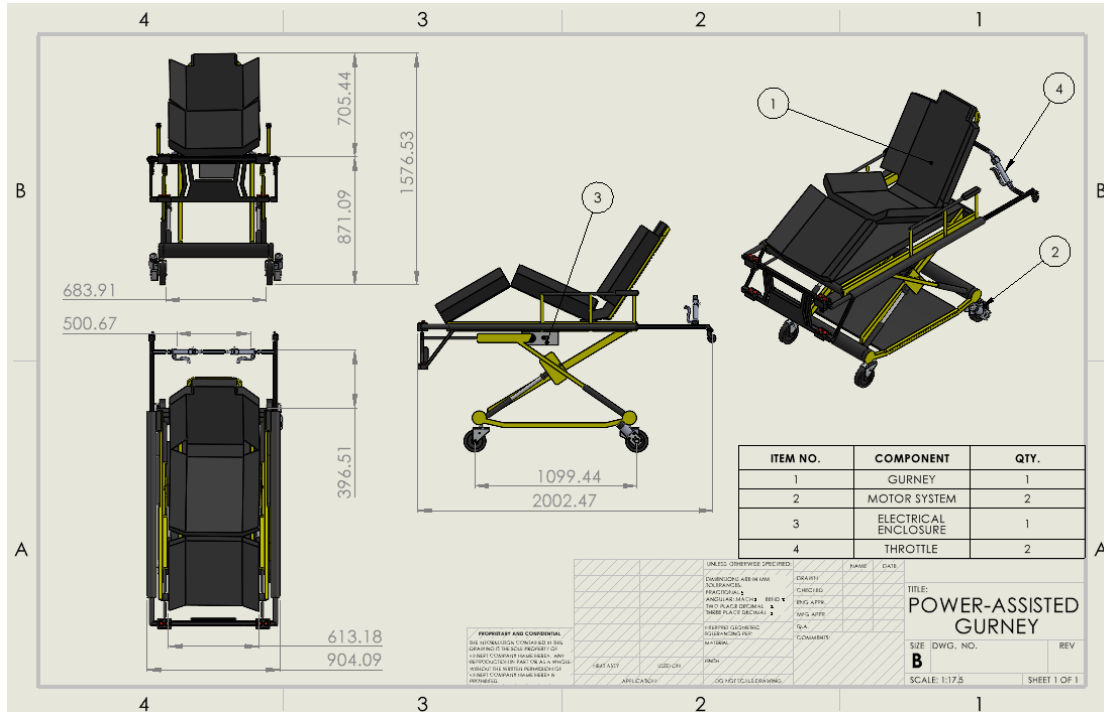


Figure 2: Power Assisted Gurney BOM Drawing.

Overall, the product is a simple feedback system that varies the power output based on the patient's weight on the gurney and the amount of control from the operator. The Power Assisted Gurney will mimic a standard gurney used by paramedics but with added features. It is designed to be used and operated like that of a stretcher. This means similar key dimensions and similar functionality to a standard stretcher. Maintaining familiar dimensions and handling characteristics will allow paramedics to transition seamlessly to using the Power-Assisted Gurney without extensive training.

Load cells located underneath the gurney are used to measure the patient's weight on the gurney. This technology is similar to that of existing pressure sensor pads which are already in use for various applications such as XSENSOR's walkway and stance pads [32] or Performance Health's Timed Bed Sensor Pads [33]. The load cell will not only detect pressure from the patient but measure it becoming an input for the system. These sensors,

as seen in Figure 3. will provide the necessary information for the microcontroller so that it may recognize the size of the patient when the gurney is in use.

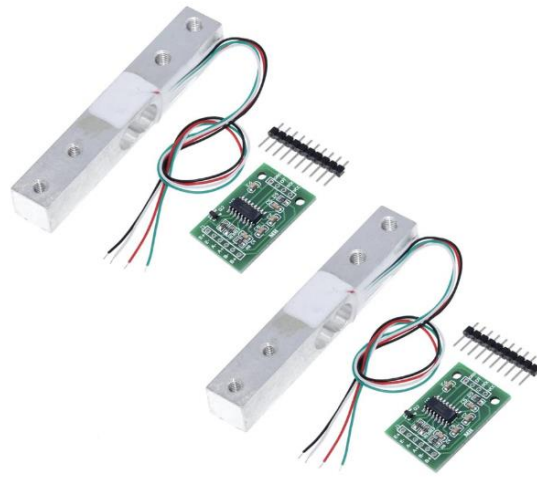


Figure 3: Power-Assisted Gurney Sensors.

To control the assisted power from the wheels, throttles are used to achieve patient maneuverability. Placed at convenient locations of the gurney's frame, operators may use these throttles to provide the desired power to either push or pull the gurney. As seen in Figure 4, they are designed to be ergonomic as they are oriented at a 45-degree angle on the frame for proper pushing posture. The design is meant to be intuitive, like using a bicycle brake handle and a simple joystick. The wheels become powered once this lever is pressed and turn off when the throttle is not in use.

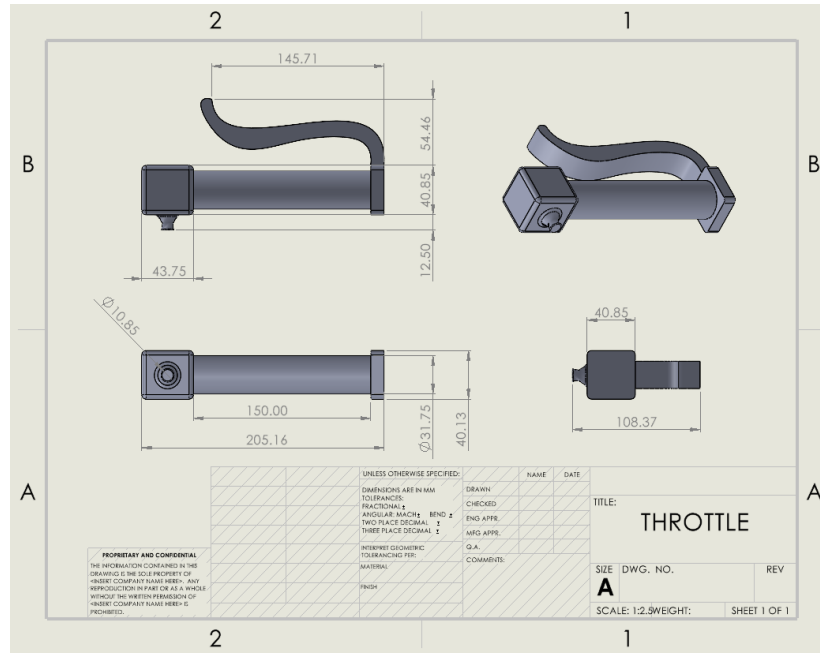


Figure 4: Power Assisted Gurney Throttle.

The STM32 microcontroller interprets the inputs from the load cells and throttles into power and velocity in real-time. The STM32, a 32-bit microcontroller referred to as the brain of the system as seen in Figure 5, will receive information and provide an output based on those inputs. In the Power-Assisted Gurney, the microcontroller will receive the measured patient's weight and the amount of desired power from the operator. With these inputs, it will calculate and provide a signal to turn on and provide power to the motorized wheels. The STM32 and its necessary accessories will be located within an electrical enclosure box under the bed of the gurney.

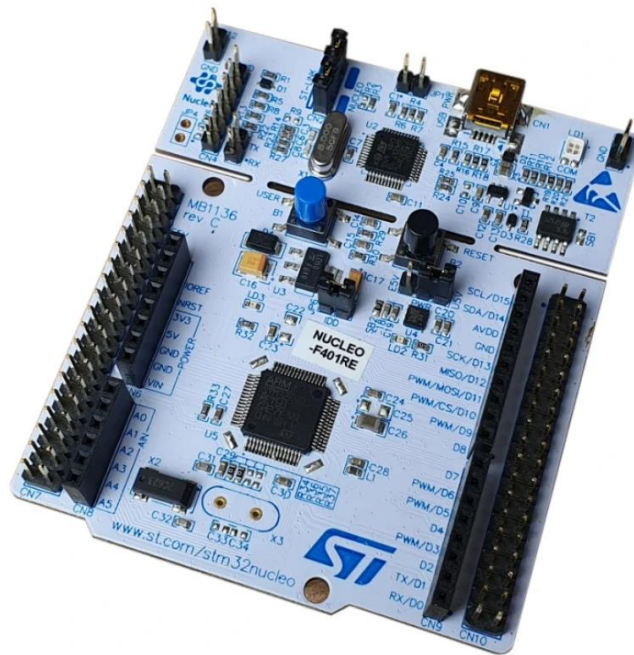


Figure 5: STM-32 Microcontroller.

Motor System

The Power-Assisted Gurney was designed to enhance patient mobility while reducing the physical burden placed on paramedics during emergency response. At the core of this system is a high-speed brushless motor selected for its compact size and high RPM capabilities. This motor is coupled with an 8:1 reduction gearbox, which was carefully chosen to amplify torque without sacrificing responsiveness or speed. Initially, the drivetrain design featured a traditional horizontal spur gear arrangement. However, to minimize the bulk of the mechanical system and reduce interference with the gurney frame and patient space, we transitioned to a right-angle worm gearbox. This refinement allowed for a more compact layout and added the benefit of natural back drive resistance, preventing unintended rollback when the gurney is stopped on an incline.

The motor system, seen in Figure 6, achieves a top speed of approximately 11km/h — sufficient for emergency mobility while maintaining control and safety. Both motors are powered simultaneously through a 24V battery system and operate in parallel, drawing a combined peak current of approximately 130.5A under maximum load. Supporting electronics are integrated into a custom-designed power distribution PCB. This board includes a buck converter to step down the 24V main battery voltage to the 3.3V and 5V logic levels required by the STM32 microcontroller and associated peripherals. Fuses are

incorporated into both high- and low-voltage circuits to provide overcurrent protection, ensuring the safe operation of all electronics.

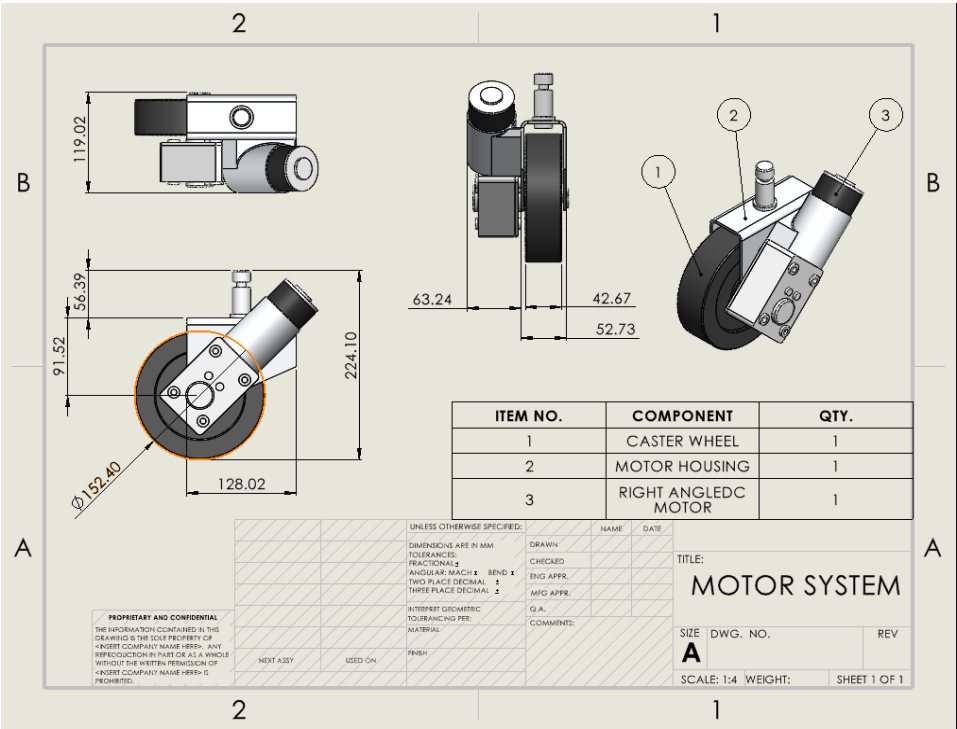


Figure 6: Power Assisted Gurney Motor System.

Microcontroller System

Designing the system for the microcontroller unit (MCU) to calculate the gurney’s velocity is essential for responsiveness during emergency scenarios and providing the correct power to the motors. The system designed (Figure 7) uses load cells to monitor patient weight, a throttle for acceleration, a single-axis joystick for steering, and an accelerometer to determine the direction of travel of the gurney.

The load cells use the HX711 amplifier for reliable communication with the MCU [34], four of the load cells are positioned under the gurney bed as shown in Figure 8 to accurately measure the weight of the patient. The throttle has an input range of [0, 1], where 0 represents no throttle and 1 represents full throttle, the user will be able to press/release the throttle until the desired speed is reached. The joystick is only able to be moved in the horizontal axis of direction with an input range of [-1, 1], where -1 represents a full left turn and 1 a full right turn. This enables the user to easily adjust the direction of travel by moving the joystick. Finally, the integrated accelerometer informs the MCU the direction of travel of the gurney in real-time with -1, 0, and 1 as the possible values. An acceleration of -1

indicates the gurney is moving backwards, 0 indicates no movement, and 1 indicates that the gurney is moving forward. The value of the accelerometer will indicate the direction the motors apply force and assist in transporting the patient.

For the MCU for the system, an STM32 microcontroller was chosen for real-time calculations due to its high performance, low power consumption, and availability [35]. The STM32, shown in Figure 9, will perform the calculations for velocity and power of the motors based on user input in real-time. The MCU will communicate these values to the motor system via the UART communication protocol which uses two communication lines for efficient data transfer [36]. Figure 10 shows a high-level view of the MCU interface with the motor system.

To ensure the firmware developed for the MCU meets medical regulations, the team will follow the IEC 62304 standards for medical device software and the software lifecycle [37].

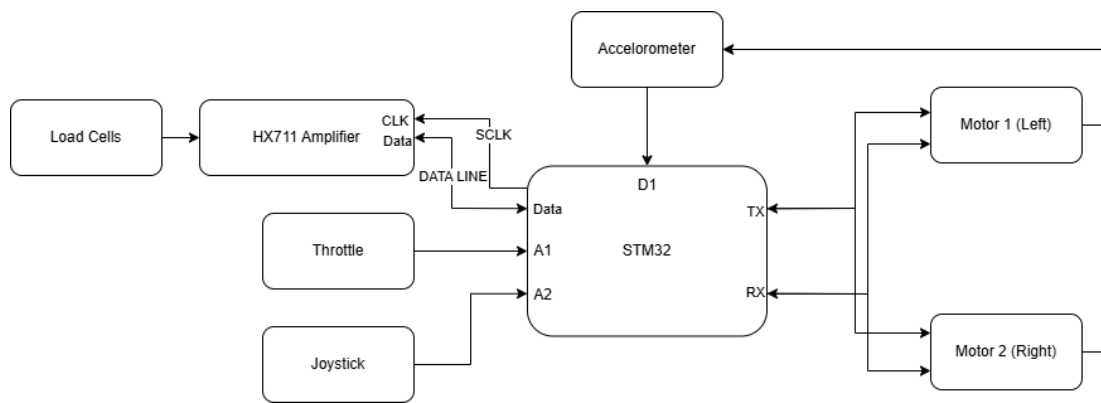


Figure 7: STM32 Microcontroller system diagram.

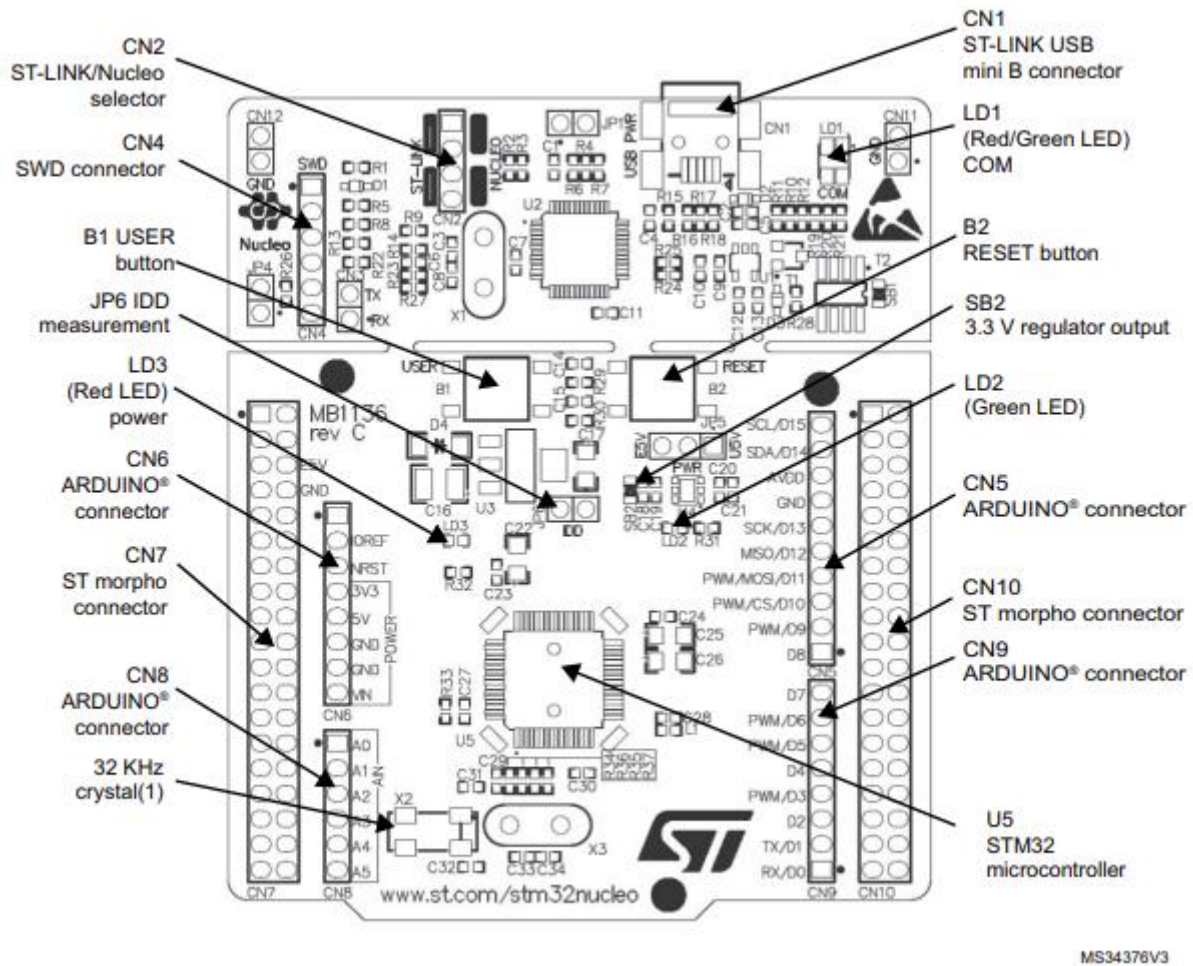


Figure 8: Top Layout of STM32 Microcontroller)

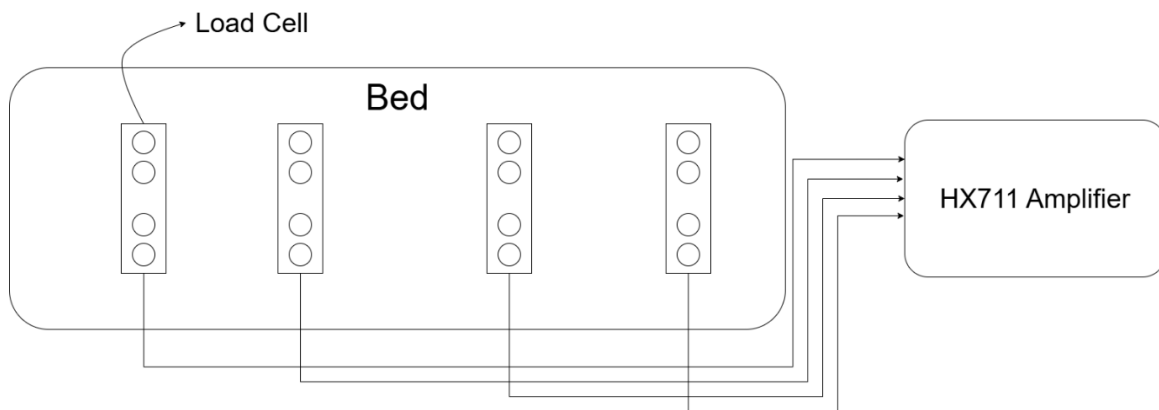


Figure 9: Load cell distribution under gurney bed.

Cost of Implementing the Design

The power assistive gurney in terms of the materials and components required to manufacture it, the total as seen in Table 6 is \$1,570.22. To begin implementing the design to a larger manufacturing scale, the first-year cost was calculated taking manufacturing and legal factors into account. The starting assumption is that within the first year, 150 gurneys would be produced. Some noted factors are expenses for staff wages, rent of a facility to hold, machinery, and the raw material cost of 150 gurneys. Additionally, the cost of training newly onboarded staff, obtaining liability insurance and regulatory certifications, the total cost to start the manufacturing of 150 gurneys in one year adds to \$958,933.

Table 6: Bill of Materials and total cost to build one gurney

Components	Model	Estimated Cost (\$)
Microcontroller	STM32	22.45
Weight Sensor Pad	XSENSOR Walkway Sensor Pad / HX711 Load Cell Kit	20.05 [38]
Motorized Wheels (x4)	Electric Hub Motors (e.g., 24V 250W Brushless Motor)	130.07
Throttle Control	Motorcycle-Style Twist Throttle	49.95
DC Motors	24V 350W High-Torque DC Motor	298.62 [39]
Battery Pack	24V 40Ah Lithium-Ion Battery	558 [40]
Motor Controller	Sabertooth Dual 25A Motor Driver	124.99 [41]
Electrical Enclosure Box	ABS Plastic Waterproof Enclosure	17.39
Wiring Connectors	JST / XT60 Connectors, 12AWG Wiring	5.99
Frame	6061-T6 Aluminum (\$5.37/kg)	365.16
Total Materials Cost (\$)		1,570.22

Table 7: Cost to implement design over 1 year assuming 150 gurneys produced

Category	Cost Breakdown	Estimated Cost (\$)
Manufacturing	Raw Materials	235,533
	Labour	300,000 [42]
	Machinery and Equipment	178,400
	Overhead (utilities, rent)	180,000
Implementation	Insurance and Legal Fees	45,000 [42]
	Training	20,000 [42]
Total Implementation Cost (\$)		958,933

DESIGN DEFENSE

FUNCTIONALITY

The Power-Assisted Gurney's intelligent system is crucial in aiding the operator in transporting the patient by either pushing or pulling while minimizing strain. According to AORN Ergonomics, it is stated that the recommended pushing/pulling force for a stretcher is 195N. With this system, it will ensure that the operator won't need to exceed that threshold no matter the weight of the patient.

With the integration of the power-assistive system on a gurney or stretcher, paramedics will no longer have to endure bodily strain when carrying out their core responsibilities. Figure 10 shows the effectiveness and impact of integrating an assistive system into a gurney or stretcher. As patients' weight increases, the force required to push both the gurney and patient also increases linearly. This indicates that paramedics will need to exceed the recommended pushing/pulling force of 195N to transport the patient. However, with an integrated motor assistance system, transporting patients becomes easier. With motor assistance, the system will engage, causing the motor to endure most of the force as patient weight increases, ensuring that paramedic force stays either below or equal to 195N. As a result, this assistive system will make sure paramedics won't experience bodily strain while transporting patients on a gurney or stretcher.

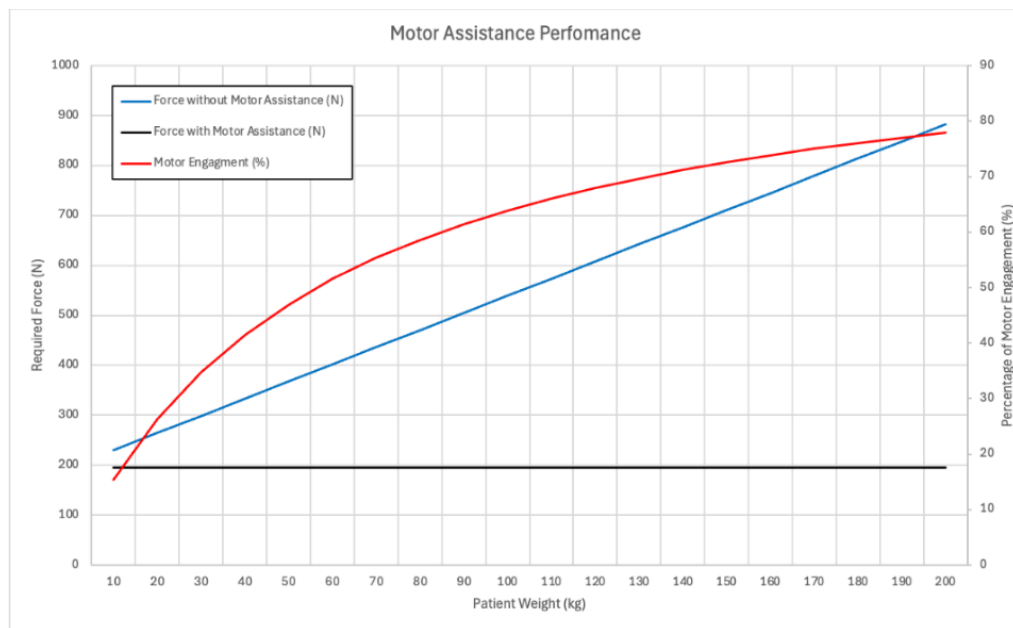


Figure 10: Motor Assistance Performance Graph.

Motor and Battery Function

To prove the system's functionality, a patient weight of 220lbs., an operator force of 195N, a minimum velocity of 10km/h, and a surface incline of 10° were used to demonstrate an extreme scenario. Using these parameters, a combination of 494.82N from the motor and 194.8N from the operator was required, resulting in a required torque of 37.98 Nm for the motor. The derived calculations can be seen in Appendix M. Due to a limit of stall torque provided by the selected motor, a gearbox was needed to increase the torque of the motor system. According to the team's torque calculation in Appendix M, a gear ratio of at least 8:1 was required to achieve 37.71Nm. With an 8:1 gearbox added to the system, the RPM decreased to 393.83RPM. However, this is still enough speed to achieve a velocity of 11.31 km/h, which is greater than the minimum required velocity of 10 km/h.

The design decisions made throughout the development of the powered gurney are grounded in both theoretical analysis and practical field considerations. The high-RPM brushless motor selected for this system produces a stall torque of 4.75 Nm. When paired with an 8:1 gearbox, the output torque increases to 37.98 Nm, which slightly exceeds the calculated threshold of 37.71 Nm — the torque required to mobilize a 220lbs. patient up a slope. Meeting and marginally surpassing this torque requirement ensures reliable traction and load-handling performance, especially in emergency scenarios involving heavier patients or inclined terrain.

The drivetrain was originally designed using a horizontal spur gear configuration, but during refinement, the team transitioned to a right-angle worm gearbox with the same 8:1 ratio. This decision was made to reduce the mechanical footprint and improve packaging within the gurney frame. In addition to minimizing visible bulk and improving aesthetics, the worm gear inherently provides back-drive resistance, offering an added layer of safety by preventing unintentional rolling on inclines.

To validate battery performance, a detailed power analysis was conducted. Under peak load, each motor draws approximately 65.25A, totaling 130.5A when both motors operate in parallel. A theoretical battery capacity was calculated in Appendix M which was 78.31Ah. This would support an hour of continuous operation at maximum load. However, by factoring in a realistic usage profile—60% at full power and 40% at half power—we calculated a more accurate runtime of approximately 45 minutes and 36 seconds as seen in Appendix M. This comfortably exceeds the average 24-minute patient contact time cited by Toronto EMS [43], ensuring consistent performance over multiple uses before requiring a recharge.

Electrical safety was another key focus of our design. The STM32 microcontroller and peripheral components receive power through a custom power distribution PCB. A buck converter steps the main 24V battery voltage down to 3.3V and 5V logic levels to prevent damage to low-voltage components. Additionally, fuses are installed at key points on both the high- and low-voltage circuits to guard against overcurrent conditions and short circuits. These protective elements ensure the reliability and safety of the entire control system during real-world operation.

In summary, our integration of a high-torque motor system, space-efficient mechanical layout, accurate power calculations, and protective circuitry demonstrates a well-justified, functional, and safe solution tailored to the dynamic and demanding needs of emergency medical responders.

STM32 Function

The calculations are performed in the MCU to translate the inputs from the user and determine the desired power and velocity for the motor system. It is essential to update the power and velocity in real-time, since a delayed response from the system could risk the transport of patients and make operating the gurney difficult.

The equations will be integrated into the STM32 firmware to process input signals and calculate the required velocity and power for each individual motor based on the input of the operator. Figure 11 illustrates the MCU configuration, including its primary inputs and outputs. A pseudocode implementation (Appendix N) was developed to read input values via analog pins and the I2C interface, and to send the corresponding output commands to the motors [44].

The derivations for velocity, angular velocity, and related constants are provided in Appendix M. A comprehensive list of assumptions, initial conditions, governing engineering laws and principles, as well as applicable codes, standards, and regulations—with justifications for their use in these calculations—is available in Appendix M.

An extreme scenario is assumed in the calculations where the operator needs to travel at the max speed of 10km/h while doing a full turn to either side. A full turn to the right is set in the chosen scenario to find the max angular velocity of each wheel, this is crucial to set a safe speed limit for each wheel while meeting the power and velocity demands of the user. The calculations in Appendix M indicate that the max thresholds for speed, angular velocity, and acceleration of the wheels are 2.78m/s, 9.59rad/s, and 6.30m/s² respectively.

With the calculated thresholds for the max speed, max angular velocity, and acceleration; system limits must be integrated into the STM32 firmware. Implementing safe thresholds is critical for ensuring safety and reliability by preventing the system from requesting a power value that could compromise patient safety, delivering excessive power that could damage the motors, and cause unintended behaviors during operation.

The main source of error that could negatively affect the system is the noise coming from the input modules. Specifically, the load cells can experience noise when exposed to radio or electromagnetic signals [45]. These signals are common in ambulances, making it a significant source of error that needs to be addressed. A moving average filter will be implemented into the system to filter the input from the HX711 amplifier module. This filter is an ideal choice to smooth out short-term fluctuations in the incoming signal, providing a more accurate weight reading [46].

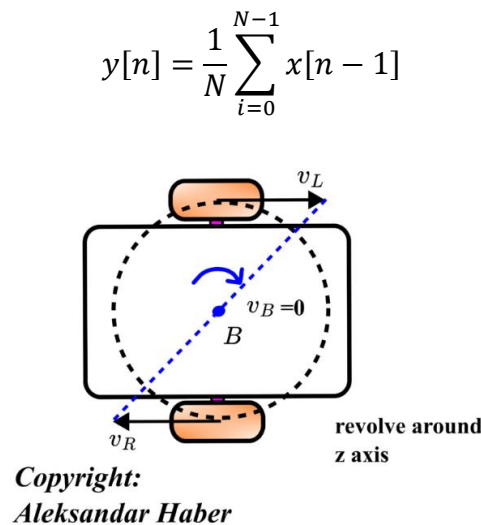


Figure 11: Diagram for angular velocity calculation in a revolution around Z axis.

Safety

Our design adheres to the weight and dimension tolerances offered in the Provincial Equipment Standards for Ontario Ambulance Services (O. Reg 257/00), listed in Appendix D. Our main safety features are listed Table 8 next to the standards for which they are relevant.

Table 8: Main Safety Features

EHRAB Equipment Standard No.	Relevant Safety Features
80 Cot, Lift Assist	- Our design can be shortened, and our 0-point turn features allow for turning in tight areas.

	- Lockable side rails allow for further patient support. - Can support the weight of, at minimum, a 90th percentile male.
300 Restraining Straps, Cot	- Our design is equipped with a Patient Securing System (PSS) (4 straps). - The PSS can support 10x the weight of a 90th percentile male.

Furthermore, the system can be safely stored and restrained in an ambulance, to prevent loose cargo. All patient-adjacent materials can be disinfected, and any metal frame components are painted to prevent rust.

Additional safety features include an emergency stop (E-Stop). Standard ISO13850 [47] was followed in the design and review of our E-Stop actuator, nameplate, location and protective shielding. We also have lock-out tag-out features for any maintenance work to be done on the electronics.

Finally, our system offers power-assisted braking if it is required during the traversal of inclined terrain.

ECONOMIC

Canadian hospitals allocate 5.4% of their yearly budget on medical equipment and 63.3% on worker compensation [48]. Included within that 63.3% is recompense for those who experience injuries while performing their duties. This covers monetary compensation for pay loss while injured, rehabilitation, and a pension if the injury results in permanent disability. When it comes to duties that paramedics perform daily, this can result in tens of thousands of dollars spent aiding the injured when incidents can be avoided. The impact of MSD incidences leads to an increase in lost workdays costs upwards to \$74,000 a year as injury rates increase [49]. However, with the introduction of the power assistive gurney, it is shown that the rates of injury can be decreased by 78% [49]. This reduction in risk brings economic savings to what are avoidable injuries to paramedics as fewer days of work will be missed and less can be spent on injury compensation.

The expenses on equipment can also be saved with the power assistive gurney when compared to competitors. In 2022, the Niagara Emergency Medical Services spent a total of \$2,451,919 on 50 Stryker Power Pro-XT stretchers, making each a price of \$49,038.38 [20]. Niagara Medical Services stated that the costs to retrofit ambulances with an alternate stretcher and to train the staff would be \$800,000 and \$80,000 respectively [20]. In contrast, AMBUTECH's power assistive gurney costs \$10,000 per unit, costing a total of \$500,000 if 50 were purchased. Even in a maximum expense scenario, the total cost would be an

estimated \$1,380,000 for 50 power assistive gurneys from AMBUTECH. This makes it nearly half the cost compared to purchasing from a competitor.

SOCIAL/ENVIRONMENTAL

The power assistive gurney can make great improvements to the lives of patients and paramedics. With the introduction of AMBUTECH's gurney solution, it can be expected that injuries among paramedics can be reduced by as much as 78% [20]. This reduction allowed for our proposed motorized gurney increases the quality of life for paramedics as the risk of developing a MSD at work dramatically decreases, allowing for more fulfilling long-term careers, and a future with no chronic pain.

With the risk of injury and the work performed lowered, the time it takes to get injured patients to the hospital can be quicker. Efficiency gains can be seen as with more work being done by the motors, less personnel would be essential for moving the gurney around in the field or in the hospital, freeing more hands to tend to alternative tasks in critical situations [50]. Decreased manpower makes it safer for the paramedics and nurses to get the patient to where they need to be heading. This benefits not only the quality of life for the paramedics and nurses, but also the patient who will be receiving the care they need sooner.

The manufacturing and use of technical medical equipment accounts for 4.4% of the global carbon emissions [3]. To ensure that the power assistive gurney has as low of a carbon footprint as possible, the materials used in manufacturing are pre-existing components. Competitors often create proprietary parts that require unique raw materials or special processing techniques, leading to increased energy consumption and emissions [51]. In comparison, utilizing sourced parts allows manufacturing to leverage pre-existing supply chains. Additionally, the use of sourced parts makes recycling of the power assistive gurney in the end-of-life phase easier to dispose or reuse, further reducing waste and emissions.

LIFE CYCLE ANALYSIS

Design Production Emissions

To calculate the total carbon dioxide emissions that would be produced from the manufacture of the power assistive gurney, the amount of carbon emitted in kilograms from each material type used in the anticipated final design was calculated to be a total of 11,782.84 kg of carbon dioxide per gurney developed. This can be seen in Table 9.

Table 9: Carbon dioxide emissions from the design of the gurney

Material	Total Weight (kg)	CO ₂ Emissions (kg)
Aluminum	68.04	1088.64 [52]
Copper	6.21	25.461 [53]
Plastic	15	37.5 [54]
Magnets	0.83	9.961 [55]
Polyurethane Foam	2.26	11.28 [56]
Total Emissions (kg)		1172.842

Transportation Emissions

To determine the total emissions released from the import of the components used in the final design of the power assistive gurney. Utilizing literature values of the amount of carbon dioxide emitted during air freight travel [57], the total transportation emissions in kilograms was determined to be 191.54kg per gurney. This can be seen Table 10.

Table 10: Carbon dioxide emissions from the transportation of components

Component	Weight (kg)	Distance Traveled (km)	CO ₂ Emissions (kg)
Microcontroller	0.068	11,756	0.52
Weight Sensor Pad	0.052	11,169	0.4
Motorized Wheels (x4)	12.4	12,517	95.18
Throttle Control	0.225	550	0.17
DC Motors	12	12,281	92.11
Battery Pack	4.3	3,968.1	0.54
Motor Controller	0.09	588.4	0.07
Electrical Enclosure Box	0.277	12,516	1.74
Wiring Connectors	0.200	761	0.28
Frame	68.04	76.5	0.53
Total Emissions (kg)			191.54

Operations Emissions

The amount of carbon dioxide emitted during the operational life span of the power assistive gurney was calculated in Table 11. For these calculations, it is assumed that the gurney would be in use for 12 hours a day, 365 days a year for a total life span of 7 years. Calculating for the lifetime power usage in kilowatt hours and converting to the amount of

carbon dioxide emissions in Canada [58], the total emissions for the power assistive gurney operating over a 7-year life span was determined to be 8,664.69 kg of carbon dioxide.

Table 11: Carbon dioxide emissions from gurney operations

Component	Power (W)	Lifetime Power (kWh)	CO ₂ Emissions (kg)
Microcontroller	0.003438	0.1054	0.017918
DC Motors	350	10,731	1,824.27
Battery Pack	300	9,198	1,565.66
Motor Controller	12	367.92	62.5464
Motorized Wheels	1000	30,660	5,212.20
Total Emissions (kg)			8,664.69

End-of-Life Emissions

When the time comes for the power assistive gurney to be taken out of commission and recycled, the amount of carbon dioxide produced from 1kg of waste, is 700g of carbon dioxide [59]. This makes the emissions of the gurney which holds a weight of 97.65kg, to be 68.36 kg of carbon dioxide.

RISKS AND UNCERTAINTIES

As with any engineering project, the design and implementation of the power-assisted gurney involve several risks and uncertainties that must be identified and addressed to ensure its reliability, safety, and usability.

One of the primary risks identified is battery reliability. Since the gurney relies on electrical power to assist paramedics during patient transport, a failure in the battery system could compromise its functionality, requiring users to revert to manual operation. To mitigate this, the team has selected a high-capacity battery that can support extended use during typical EMS deployments. Another important risk is durability and maintenance. Gurneys used in emergency medical environments are subjected to frequent and intense use. This includes exposure to rough terrain, high patient turnover, and time-sensitive conditions. To address this, the gurney has been designed using robust components and materials selected for their strength and reliability under stress. Maintenance has also been considered during design to allow for easy access and servicing of key components.

In terms of uncertainties, one major challenge is predicting how the gurney will perform in real-world emergency conditions, including varied terrain, weather, and time-sensitive operations. These unpredictable factors may affect the performance of the motorized systems. To account for this, a sensitivity analysis has been conducted, and further real-world testing is planned to validate and refine the design under operational conditions. Another uncertainty involves user adaptation and training. Since the power-assisted gurney introduces new control interfaces, there is potential for a learning curve among paramedics who are accustomed to manual operation. To minimize this, the design incorporates intuitive, user-friendly controls, and in the future, paramedic feedback will be continuously collected throughout testing to improve the usability and ensure smooth adoption into practice. By identifying and addressing these risks and uncertainties early in the design process, the team aims to create a solution that is not only functional but also dependable and aligned with the needs of EMS personnel.

While the current design of the power-assisted gurney addresses several present-day risks, additional challenges may arise as the product moves toward prototyping, testing, and eventual deployment. Identifying these early can help guide long-term planning and system refinement. One key future risk is long-term reliability and component wear. Over time, motors, load cells, and control systems may degrade due to repeated use, exposure to environmental elements, or mechanical stress. Even with high-quality components, real-world use across different hospitals and terrains could reveal failure points not anticipated during theoretical modelling. To address this, extended durability testing and the

development of a maintenance and replacement schedule will be essential to ensure consistent performance over the gurney's intended lifespan. Another anticipated uncertainty is integration into diverse EMS systems. While the gurney is designed with standardized ambulance dimensions and workflows in mind, regional differences in EMS practices, protocols, or equipment layouts could affect compatibility. These variations may lead to unforeseen adoption barriers. The following graph presents the Risk Assessment Matrix for the Power Assisted Gurney.

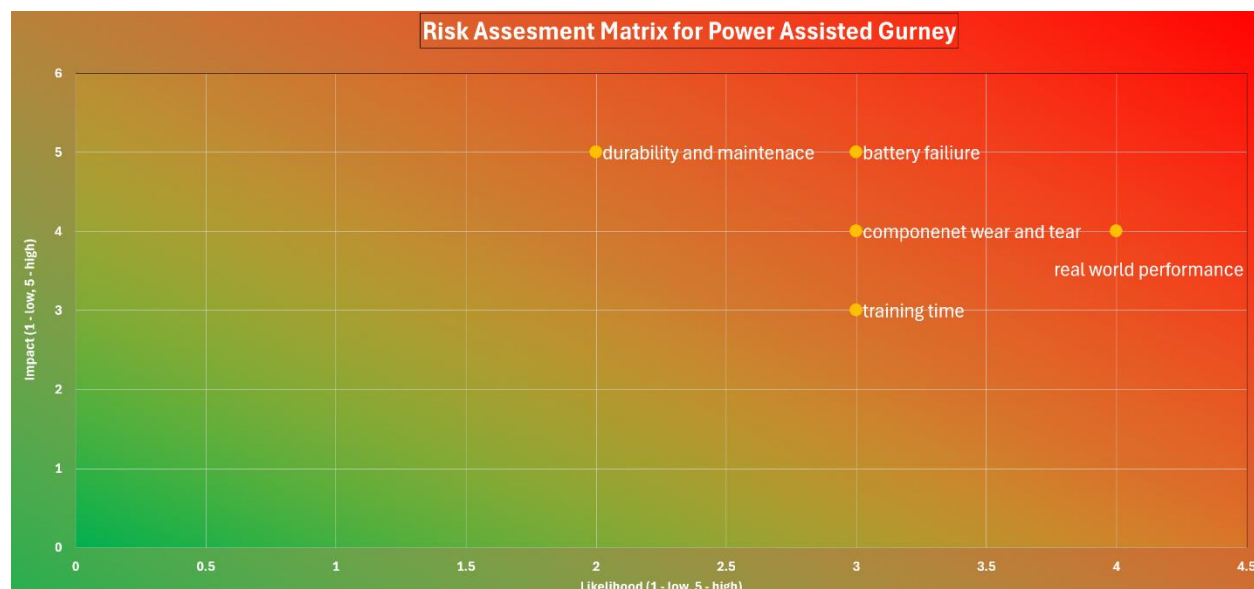


Figure 12: Risk Assessment Matrix for the Power Assisted Gurney.

PROJECT MANAGEMENT

CPM

The following outlines the Critical Path Method (CPM) outlining. The Critical Path Method determines how long a project can take and can help determine the float time. Furthermore, a CPM shows where and how much flexibility exists during the timeline of the project, helping identify the longest and shortest possible times to complete the project. Each activity is listed on a table, then used to construct a diagram, showing the logical order of the activities. Our CPM does not have a float time, earliest event time or latest event time as the diagram is linear. Table 9 and Figure 13 present the CPM.

Table 12: Critical Path Method Table

	Activity	Immediate Predecessor	Duration (days)
A	Problem Identification	None	7
B	Idea generation	Problem Identification	14
C	Design Selection	Idea Generation	3
D	Build Theoretical Model	Design Selection	7
E	Test and Evaluate with Simulations	Build Theoretical Model	14
F	Optimize design	Test and Evaluate with Simulations	7
G	Risks & Uncertainties	Optimize design	7
H	Prototype Development	Risks & Uncertainties	30
I	Real – World Testing	Prototype Development	30
J	User Feedback	Real – World Testing	7
K	Product Improvement	User Feedback	7
L	Final Design Documentation	Product Improvement	7
M	Product Deployment	Final Design Documentation	N/A

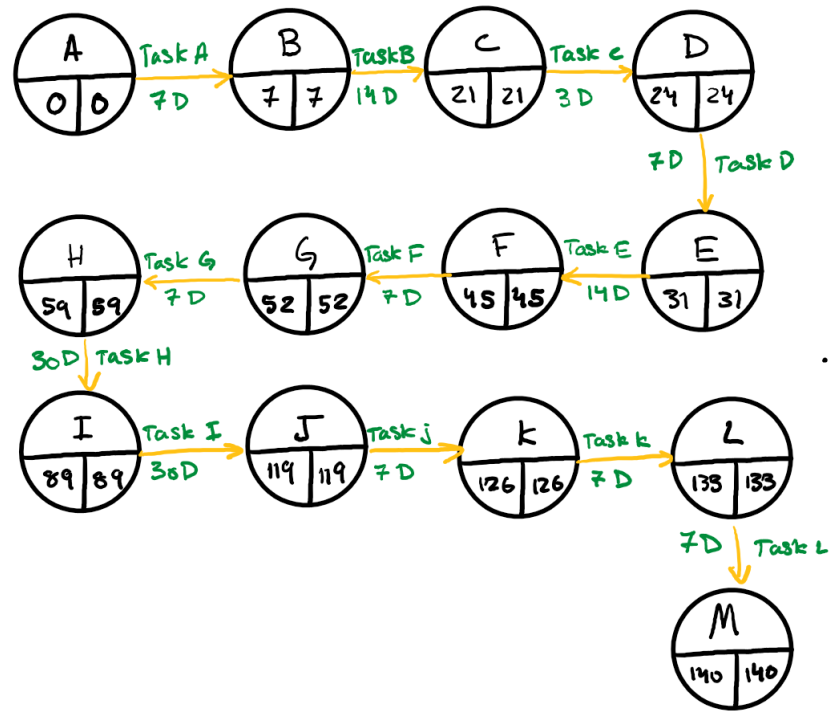


Figure 13: Critical Path Method Diagram.

GANTT CHART

The following Gantt chart outlines the project deliverables moving forward, from testing and evaluating the design using simulation to the final stage of selling the product to hospitals. The deliverables have been broken down into subtasks, broken down by week to showcase the deadlines. Figure 14 presents the Gantt chart.

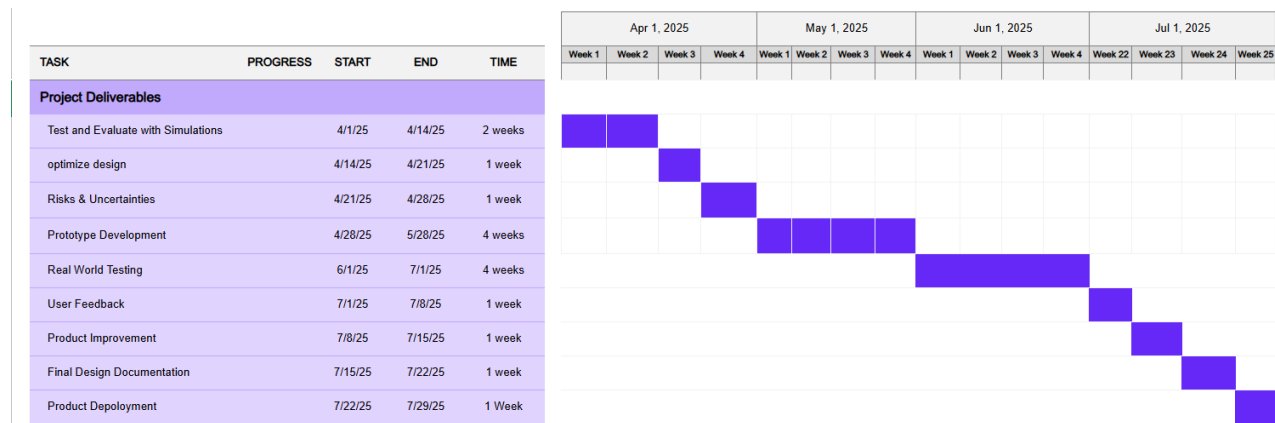


Figure 14: Gantt Chart.

ANTICIPATED CHALLENGES

One of the challenges the team faced during the development of the gurney was the pricing. During AMBUTECH's research, there were various sources online and comparable products like the Power-Assisted Gurney in the market. However, many of these sources lacked detailed breakdowns of the prices for materials, labour, manufacturing, and other expenses. To tackle this problem, the team conducted research on the costs for the required materials/parts to obtain an accurate estimate of the assembly price of the gurney.

When it comes to promoting the product, the marketing of the gurney was identified as a potential challenge. Most of the team members were unfamiliar with the market for medical equipment and this could lead to further challenges when attempting to sell the gurney. To overcome this issue, it's essential to acknowledge the team's limited experience with the medical equipment market. A practical approach would involve engaging with marketing professionals who specialize in this field, conducting thorough market research, and developing targeted strategies that clearly communicate the gurney's benefits to healthcare providers. This would help ensure that promotional efforts are both effective and well-informed by industry insights.

Another anticipated challenge is the optimization of the prototype in future stages of development. To mitigate difficulties, the team can adopt an iterative development process to continuously refine our prototype by testing and improving in small, but manageable steps. Conducting experiments that allow participants to utilize the Power-Assisted Gurney in simulated real-life scenarios is beneficial in integrating user feedback from paramedics and industry stakeholders as it is critical and provides the team with valuable insights into how the product will perform. On top of these measures, the use of simulation software, such as ANSYS for Finite Element Analysis, will also enable the team to gain insights into the structural integrity of the product.

UNANTICIPATED CHALLENGES

A crucial unanticipated challenge during the development of our project would be gaps in information and incomplete data. To mitigate this, it's important to adopt a multifaceted approach. We can implement data collection protocols to capture necessary information such as feedback and simulation results. Additionally, it is important to consult external professionals to discuss the current plan and gain insight into possible gaps in our knowledge.

More unanticipated challenges include changes in market demand and competition against our product. Mitigating changes in market demand and competition requires a proactive and flexible strategy. Continuous market research and analyzing our competition are essential to understand new trends and shifts in customer needs. By investing in product innovation, diversifying offerings, and refining the value proposition, the business can stay relevant and responsive. Additionally, implementing robust feedback loops and agile strategic planning will enable rapid adjustments to changing market conditions, helping to maintain a competitive edge.

CONCLUSIONS AND RECOMMENDATIONS

CONCLUSIONS

This final report evaluates the final design and system requirements for the powered-assistive gurney. The goal was to create an ergonomic solution that reduces musculoskeletal injuries (MSDs) during patient transport involving pushing and pulling. Using weighted decision matrices and sensitivity analyses, we identified the most suitable design by prioritizing factors such as ergonomic benefits, cost-effectiveness, and ease of integration into EMS workflows.

The team reviewed the previous interim report and technical memo to identify the crucial changes leading to the final design. Previously, the interim emphasized on evaluating the different design alternatives and choose the best design based on the team's analysis. The solution chosen was the powered-assistive Gurney to help paramedics and address ergonomic issues associated with pushing and pulling activities during patient transport. Where these issues were previously recognized as significant contributors to paramedic injuries during the proposal stage.

This design aims to reduce physical strain on paramedics, improve workplace safety, and enhance patient transport efficiency. The final stage of the design combines more thorough research on user needs and market demands, changes based on feedback from previous design stages, and adjustments in the design and calculations related to the gurney.

The power-assistive gurney integrates motorized assistance with intuitive controls and reliable safety features to comply with provincial and federal standards. While the design introduces new components and potential challenges such as battery reliability and maintenance, the benefits in reduced injury rates, improved operational efficiency, and cost savings for healthcare providers support its adoption.

RECOMMENDED ACTIONS

Moving forward in the design process, further prototyping and real-world testing will be key to improving the durability and adoption of the product. Continuous feedback from EMS professionals will guide iterative improvements, ensuring that the final product meets the demands of emergency services. Finally, further testing on battery life and durability of the gurney will be important to adapting the design to the responsibilities of a paramedic. The power-assistive gurney represents a promising advancement in the ergonomics of

medical equipment, with great potential to ensure both the safety and efficiency of patient transport for paramedics.

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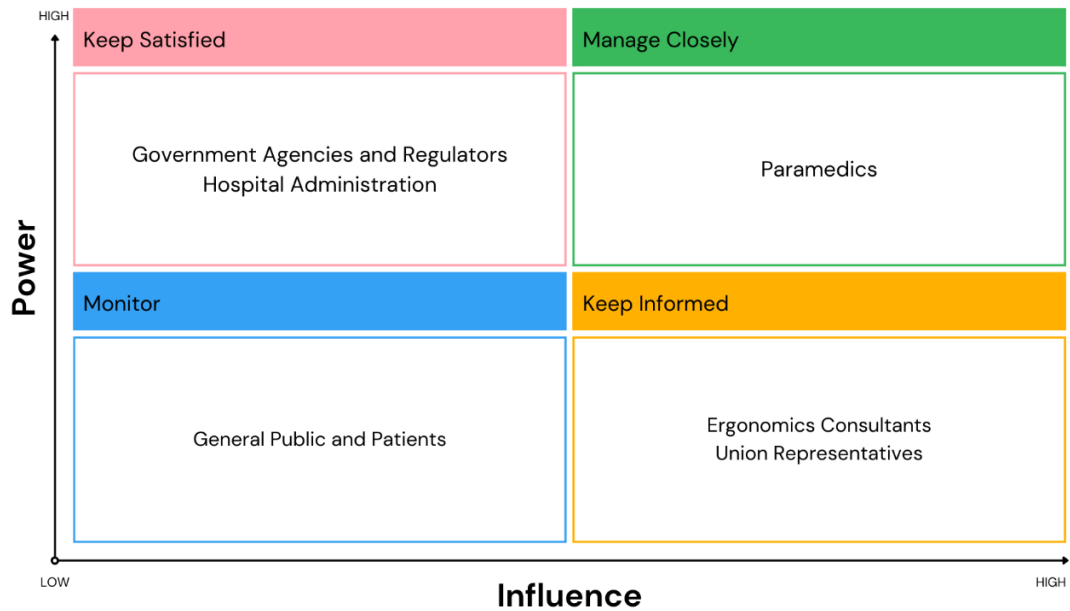
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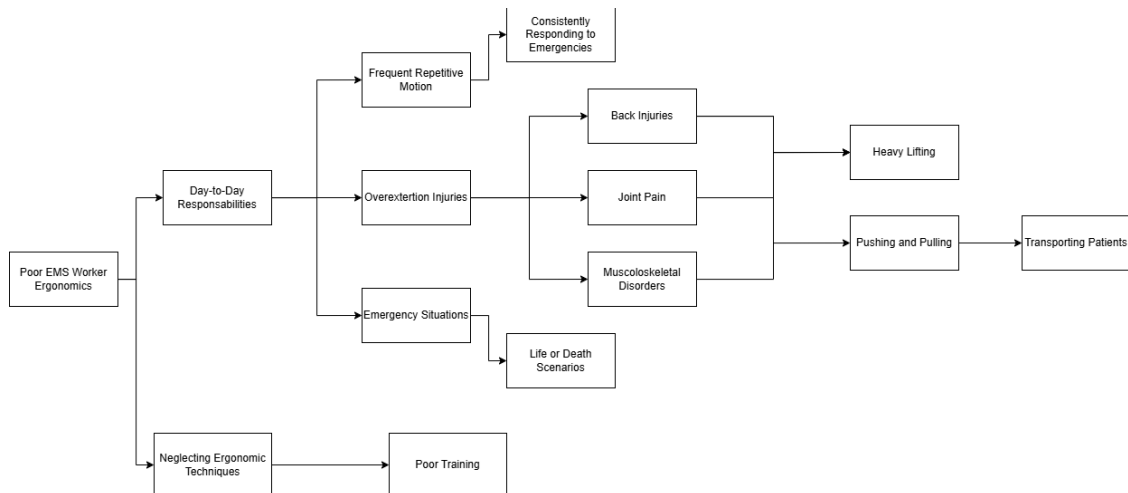
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APPENDIX

Appendix A: Stakeholder Chart



Appendix B: WHY-WHY Diagram



Appendix C: Recommended Upper Force Limits for Horizontal Pushing and Pulling

Condition	Forces that should not be exceeded, in Newtons (lbf, kgf)**	Examples of Activities
A. Standing		
1. Whole body involved	225 N (50 lbf or 23 kgf)	Truck and cart handling. Moving equipment on wheels or casters. Sliding rolls on shafts.
2. Primary arm and shoulder muscles, arms fully extended	110 N (24 lbf or 11 kgf)	Leaning over an obstacle to move an object. Pushing an object at or above shoulder height.
B. Kneeling	188 N (42 lbf or 21 kgf)	Removing or replacing a component from equipment as in maintenance work. Handling in confined work areas such as tunnels or large conduits.
C. Seated	130 N (29 lbf or 13 kgf)	Operating a vertical lever, such as a floor shift on heavy equipment. Moving trays or a product on and off conveyors.

Appendix D: Provincial Equipment Standards for Ontario Ambulance Services

No. 080 – Cot, Lift Assist	No. 165 – Infant Restraint Device
Minimum Requirements: 1. have the ability to be shortened as to facilitate its use in confined areas (e.g. elevators, tight corners, etc); 2. have an adjustable back rest that allows patient positioning from supine to fully sitting; 3. have lockable side rails that can be positioned in such a way as to facilitate the safe movement of a patient on or off the cot; 4. be manufactured with materials that can be easily cleaned and disinfected; 5. be compatible with a cot retention system that is compliant with the performance requirements contained in the current version of the <i>Ontario Provincial Land Ambulance and Emergency Response Vehicle Standard</i>; 6. have at least four (4) omni-directional wheels with solid rubber or equivalent tires; wheels mounted to ensure balanced weight distribution; 7. be equipped with a patient securing system (MOH Standard #300); 8. have an integrated, or removable intravenous pole that securely locks in place when in the upright position;	Minimum Requirements: 1. be easy to clean and disinfect; 2. be constructed with one-piece webbing straps with quick-release buckles which securely attach the device to the ambulance cot frames at the minimum of three points; 3. be constructed with a five-point harness system that consists of shoulder, chest, and crotch restraints that secure with a single, quick release buckle. All restraints shall be fully adjustable; 4. be able to be stored and secured safely in an ambulance; 5. be compatible for use with cots carried in the ambulance; 6. be capable of accepting additional head and body support to ensure patient security; 7. meet or exceed CSA standards

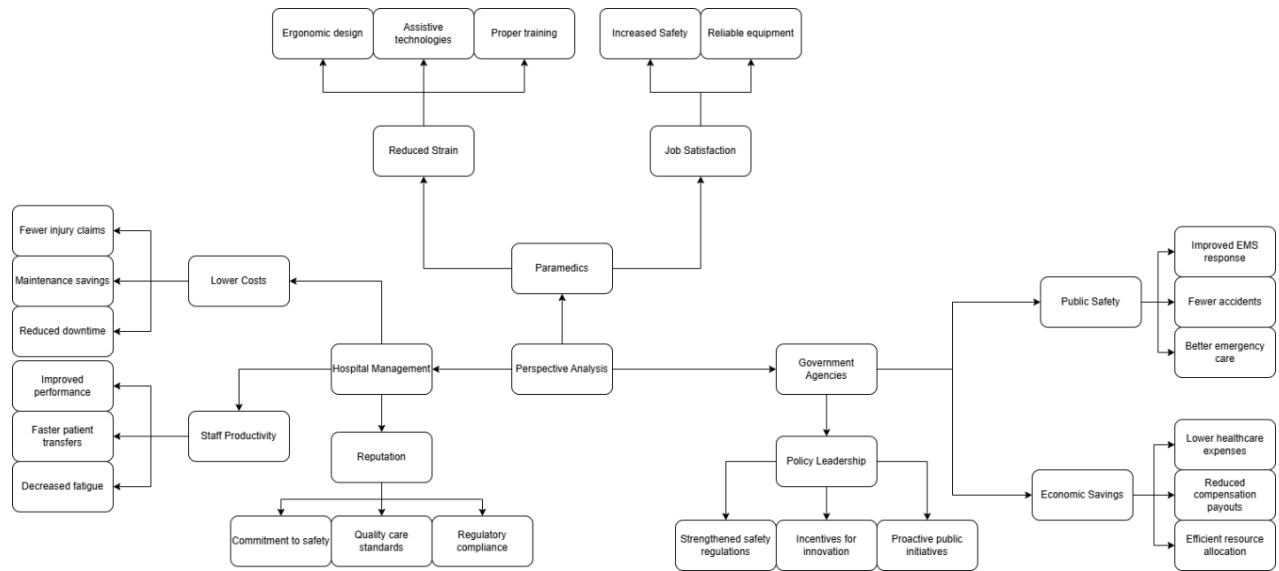
9. have the following dimensions/tolerances: - loading wheel height to be sufficient to allow cot to be loaded into all ambulances approved for use in the Province of Ontario, without manually lifting the front end of the cot off the ground; - be capable of carrying, as a minimum, the weight of a 90th percentile male; - physical dimensions of the cot bed to accommodate, as a minimum, a 90th percentile male.	
No. 205 – Lifting Chair	No. 300 – Restraining Straps, Cot
Minimum Requirements: - be manufactured with materials that can be easily cleaned and disinfected; - have non-slip handgrips on all carrying handles; have wheels, or tracks, or other method to roll the device; - have patient securing straps (MOH Standard #210) - have the following dimensions/tolerances: - be capable of carrying, as a minimum, the weight of a 90th percentile male; - physical dimensions of the lifting chair to accommodate, as a minimum, a 90th percentile male.	Minimum Requirements: - consist of two (2) complete two (2) piece "Tang & Receiver" assemblies, li.e. two (2) shoulder, one (1) waist, and one (1) leg]; - be constructed of materials that are easily cleaned and disinfected, minimum two (2) inches in width; - employ the automotive style "Tang & Receiver" fastening device configuration (quick release press mechanism); - have a minimum extended Tang and strap length of 114 cm (45 inches); - have a minimum extended Receiver and strap length of 38 cm (15 inches); be capable, as a system, of restraining, as a minimum, ten (10) times the weight of a 90th percentile male.

No. 305 – Restraining Straps, Adjustable Break-Away Stretcher & Spinal Board	No. 345 – Stretcher, Adjustable Break-Away
Minimum Requirements: 1. consist of two (2) complete two (2) piece "Tang & Receiver" assemblies, li.e. two (2) shoulder, one (1) waist and one (1) leg]; 2. be constructed of materials that are easily cleaned and disinfected, minimum two (2) inches in width; 3. have a minimum extended Tang and strap length of 122 cm (48 inches); 4. have a minimum extended Receiver and strap length of 33 cm (13 inches); 5. be capable of restraining, as a minimum, the weight of a 90th percentile male; 6. have quick release hooks that: a. are corrosion resistant and have a spring loaded snap closure; b. fit over a maximum 10 mm (3/8 inch) rod; c. have a 360° swivel capability.	Minimum Requirements: 1. be adjustable to accommodate patients of various sizes; 2. be equipped with quick connect pins with an individual pull rating of 136 kg (300 lbs) allowing the use of restraining straps (MOH Standard #305); 3. be capable of being safely stored in an ambulance; 4. have a surface that cradles the patient providing support and minimizing lateral movement; 5. be capable of carrying, as a minimum, the weight of a 90th percentile male; 6. be manufactured with materials that can be easily cleaned and disinfected; 7. have the following dimensions/tolerances: a. the physical dimensions of the adjustable break-away stretcher to accommodate, as a minimum, a 90th percentile male.
No. 355 – Stretcher, Portable	
Minimum Requirements: 1. include a transport surface of approved medical grade material that can accommodate the size of the 90th percentile male and have pole channels or handgrips; 2. be capable of carrying, as a minimum, the weight of a 90th percentile male; 3. have poles constructed of minimum 2.5 cm diameter tubing	

with non-slip handles at each end
(if using poles);

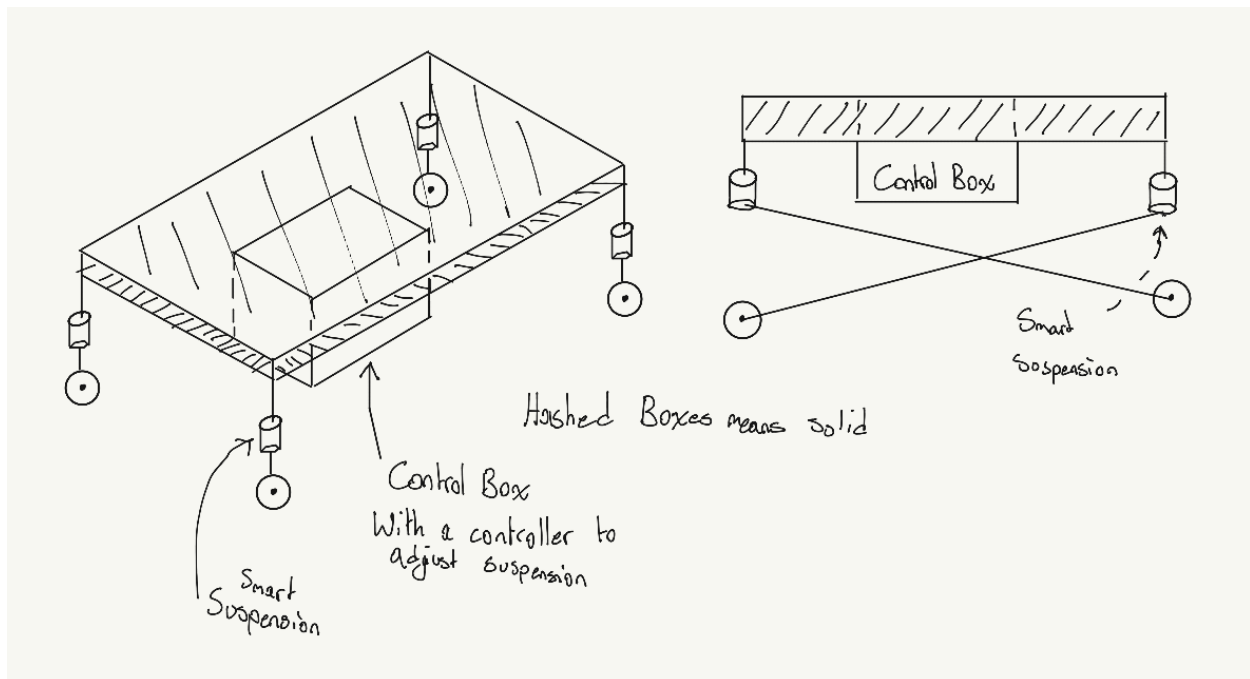
4. have pole ends with reduced diameter and spring loaded pin locking mechanism. Poles to be minimum 188 cm (74 inches) long assembled, minimum 101.5 cm long disassembled (if using poles);
5. be manufactured with materials that can be easily cleaned and disinfected;
6. have the following dimensions/tolerances:
 - a. the physical dimensions of the portable stretcher to accommodate, as a minimum, a 90th percentile male.

Appendix E: Shift Perspective Diagram



Appendix F: Alternative Solutions

Smart Suspension Gurney Concept Sketch

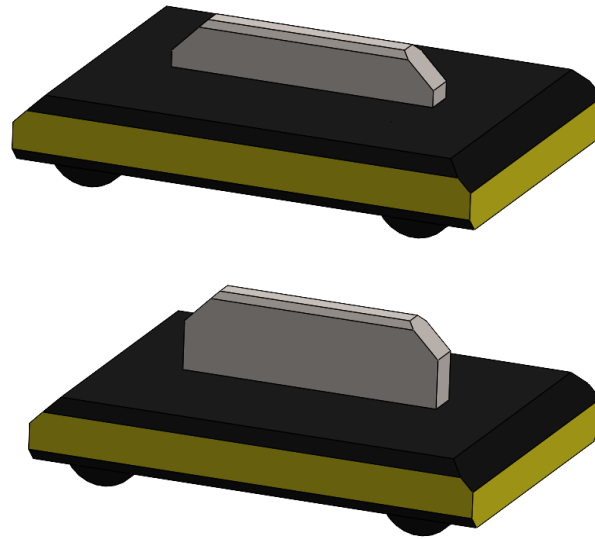


Design Constraint	Justification
Weight Capacity	The gurney supports a minimum weight of 250 lbs (113 kg) and does not introduce structural compromises that would reduce standard gurney load capacity.
Size	The gurney fits within 72" x 24" - the for a battery pack, OBC and electrical components are efficiently integrated to minimize size.
Standards Compliance	The Smart Suspension Gurney aligns with CSA Standards by incorporating safety mechanisms that improve patient handling and reduce strain on paramedics. The shock-absorbing suspension enhances stability during patient transport, exceeding safety and ergonomic standards.
Durability and Maintenance	The device is expected to last well over 5 years of use; however, the design includes multiple electronic components and sensors, increasing the likelihood of wear and tear over time. This model requires ongoing maintenance of electronic and hydraulic systems, which could impact long-term cost-effectiveness and reliability. The need for regular sensor calibration, damper checks, and battery replacements may present maintenance challenges.
Ergonomics	The device improves ergonomics – the adaptive suspension system actively reduces vibration and impact forces, ensuring that paramedics do not experience excessive strain or fatigue while maneuvering the gurney.

	Dynamic adjustments based on terrain minimize awkward postures and excessive force exertion.
Safety	The design is safe - the suspension system improves stability, preventing unexpected movements and sudden shifts in patient weight distribution, which could otherwise increase the risk of accidents. Additional locking features may be needed to ensure that the dampers remain firmly engaged when the gurney is stationary.

Design Criteria	Justification
Minimal Cost	The integration of sensors, an onboard computer, and active dampers significantly increases manufacturing and maintenance costs . The reliance on high-precision components and software-controlled systems may limit affordability for lower-budget EMS departments.
Minimal Training Requirements	The Smart Suspension Gurney does not require complex manual operation . The adjustments are automatic, allowing paramedics to use it without additional effort or major procedural changes. Training would likely be minimal, focused only on basic system troubleshooting and battery management.
Minimal Weight	The inclusion of a battery, sensors, and OBC components increases the weight of the gurney . While the adaptive suspension improves maneuverability, the added weight may increase lifting difficulty when loading the gurney into an ambulance.
Minimal Cleaning Time	The presence of electronic components introduces challenges for cleaning and disinfection. Unlike fully mechanical gurneys, electronic enclosures must be integrated to prevent damage from sanitization chemicals, moisture, or contamination.
Maximum Battery Life	The system requires continuous power, meaning battery life is crucial. While power draw is relatively low, frequent adjustments could drain battery life faster than expected .

AGV Assisted Gurney Concept Model

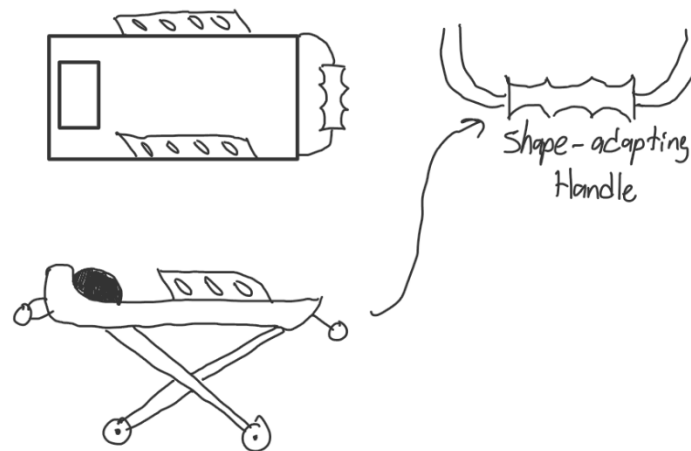


Design Constraint	Justification
Weight Capacity	This device does not directly carry the weight of the gurney entirely. Furthermore, AGVs can easily be rated for 250lbs.
Size	The device would be smaller than the footprint of already existing gurneys.
Standards Compliance	This style of device is already in use in hospital environments and would comply with standards in a similar fashion.
Durability and Maintenance	AGV battery life is approximately 3 years, with an entire AGV lifespan of 6 years [35].
Ergonomics	The device would be operated using waist-eye-height wireless controls. Bending over to access the device would only need to be done during maintenance.
Safety	The device would incorporate multiple collision-prevention and detection systems.

Design Criteria	Justification
Minimal Cost	The main cost of this product would be the large upfront cost of buying the devices and installing routes, charging stations, and control panels – maintenance and operating costs are minimal.
Minimal Training Requirements	The control panel would be intuitive and would only require an operator to understand a few basic functions, minimizing training.
Minimal Weight	The device carries its own weight in operation and when returning to charging stations. Therefore, its weight does not impact users in a meaningful way.

Minimal Cleaning Time	The device operates close to ground level, where there is no interaction with a patient. This means it wouldn't have to be cleaned meticulously.
Maximum Battery Life	The device can guide itself to charging stations when not in use, meaning that the end-user can always request a fully recharged device.

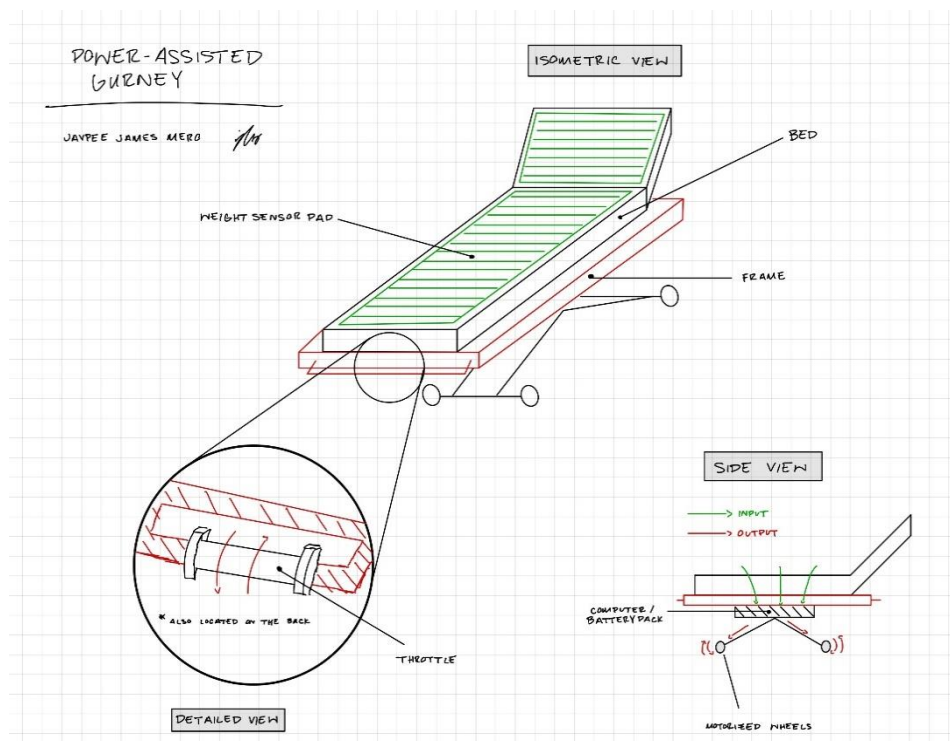
Shape-Adapting Handle Gurney Concept Sketch



Design Constraint	Justification
Weight Capacity	Similar to existing gurneys, the Adapting-Handle Gurney will be capable of safely supporting the minimum weight requirement. New features will only affect the operation of the gurney while transporting a patient.
Size	The Adapting-Handle gurney will be the same size as currently used gurneys. The new handle feature added will not make the device significantly larger.
Standards Compliance	The new gurney design is designed to adhere to the relevant government standards just like existing gurneys. The handle feature added does not violate the existing standards.
Durability and Maintenance	The Adapting-Handle gurney will be made to be used in multiple environments. The main concern would be the durability of the smart cushioning as it could start to deteriorate with constant use. This will require increased maintenance to ensure the smart cushioning stays operational.
Ergonomics	The operation of the Adapting-Handle gurney will be the same as a regular gurney. The product is meant to reduce strain of paramedics when handling the gurney.
Safety	The Adapting-Handle gurney will include the existing gurney safety mechanisms of gurneys such as breaks, straps, and sturdy frames.

Design Criteria	Justification
Minimal Cost	The addition of a specialized handle will increase the manufacturing and maintenance costs when compared to existing gurneys, however, the price of the gurney will stay lower than an electrically powered device.
Minimal Training Requirements	The Adapting-Handle gurney aims to function the same as the currently used gurneys. No extra training will be required to operate the gurney.
Minimal Weight	The device's modifications do not introduce significant weight compared to already existing gurneys.
Minimal Cleaning Time	The Adapting-Handle gurney will require more time to properly clean due to specialized handle. The extra time spent cleaning the gurney should not be significantly higher than that of existing gurneys.
Maximum Battery Life	The Adapting-Handle gurney does not rely on battery power.

Power-Assisted Gurney



Design Constraint	Justification
Weight Capacity	The addition of its features will only affect the paramedic's utilization of the gurney while safely transporting any patient – they will not reduce the gurney's standard weight capacity.

Size	The Power-Assisted Gurney can fit within standard ambulance spaces. The weight sensor is inside the bed, the computer is underneath the gurney, and the throttle is located on the frame of the gurney.
Standards Compliance	The Power-Assisted Gurney adheres to relevant standards, similar to any gurney used in ambulance service today. The addition of its features does not seem to impose any risk of violating standards.
Durability and Maintenance	The Power-Assisted Gurney can be used in various environmental conditions. Concerns include rain, snow, slush, sand, and other small debris which could affect the electronics such as the computer, motors, and throttle. Maintenance is vital to extend the life of components.
Ergonomics	The Power-Assisted Gurney will require a greatly reduced pushing/pulling force to operate.
Safety	The Power-Assisted Gurney will return to manual mode when turned off or when battery runs out. In addition, it will also switch to manual mode when the throttle is not in use. This ensures the wheels are not powered when not in use, especially when supporting a patient.

Design Criteria	Justification
Minimal Cost	The addition of electronics will most definitely result in a higher cost. Integrating a computer, sensors, throttle, and motorized wheel will result in an increase of initial cost compared to that of a normal gurney, and perhaps a smaller increase compared to a powered gurney used by Stryker and Ferno.
Minimal Training Requirements	The Power-Assisted Device is supposed to be intuitive and simple for operators. Perhaps use of the computer and throttle would be the only training added for the Paramedics. A manual of utilization and theory of the Power-Assisted Gurney can also add to the training.
Minimal Weight	The device's modifications do not introduce significant weight compared to already existing gurneys.
Minimal Cleaning Time	The Power-Assisted Gurney should only require a little more cleaning time, due to electronics, compared to that of a normal gurney used in ambulance services today. Just like any gurney, the main concern for disinfecting would be the bed.
Maximum Battery Life	The Power-Assisted Gurney will use batteries similar to that of existing powered gurneys. This ensures that this design is capable of enduring similar battery life or longer to that of its competitors.

Appendix G: Criteria and Assigned Weighting

Design Criteria	Weight	Justification
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Maneuverability	35%	The ability to push and pull the gurney is the most critical factor in patient transport. Function is a critical evaluation criterion for design[60].
Durability and Maintenance	20%	EMS gurneys are used frequently and in demanding conditions - they must be able to withstand daily wear and tear, exposure to rough surfaces, and frequent disinfection procedures. The average life expectancy of a gurney is 3 to 5 years [61]. A gurney that lasts > 5 years is ideal, as it reduces replacement costs and ensures reliability in emergency situations. While durability is vital, maneuverability takes precedence as paramedics prioritize efficiency over long-term cost savings in critical situations.
Cost	15%	Cost was weighed to reflect its importance while ensuring that the functionality is not compromised by price. Gurney models are available for < \$5,000, while powered gurneys can exceed \$30,000 [62] . Given the wide range of prices, cost must be considered, but it is not the highest priority since investments in durability and ergonomics can lead to long-term savings by reducing injury risks and repair costs.
Ease of Use and Training	15%	Gurneys should be intuitive to operate and require minimal training for paramedics to use effectively. Shorter training times allow for quicker implementation, making gurneys with training times under one hour ideal. While some training is acceptable, a gurney should not be so complex that it requires excessive onboarding, which could delay its adoption and effectiveness in the field.
Battery Life	15%	Powered gurney models rely on motorized assistance for lifting and transporting patients. The Stryker SMRT Power System demonstrates that an efficient battery can charge in two hours and last an entire EMS work shift [63]. A short battery life would be unreliable in emergency situations, requiring frequent recharge and causing potential disruptions. While this factor is important, it is weighted lower than maneuverability and durability because not all gurneys rely on batteries—manual models do not require power and remain operational regardless of charging status.

Appendix H: Decision Matrix Evaluation Scale

Rating	Perfect	Very Good	Satisfactory	Tolerable	Inadequate
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Criteria	5	4	3	2	1
Maneuverability (ease of operation)	Extremely easy	Easy; some effort on uneven terrain	Moderate effort	Difficult	Very Difficult
Cost (CAD)	< 5,000	5,000 – 10,000	10,000 – 20,000	20,000 – 30,000	> 30,000
Durability and Maintenance (years of reliable use)	> 5 years	3-5 years	2-3 years	1-2 years	< 1 year
Ease of Use and Training (training hours required)	< 1 hour	1 – 2 hours	2 – 4 hours	4 – 6 hours	> 6 hours
Battery Life (hours of use)	> 16 hours	12 – 16 hours	8 – 12 hours	4 - 8 hours	< 4 hours

Appendix I: Final Weighted Score

Final Score		Justification
AGV Assisted Gurney	3.35	- Able to autonomously transport patients, reducing the physical burden on paramedics. - Require advanced sensors, mapping software, and potential facility modifications, making them an expensive solution. - AGV components require regular calibration and servicing. - Staff would need training in operation and troubleshooting. - AGV systems typically have long battery life and return to charging stations automatically when needed.
Shape Adapting Handle Gurney	3.35	- Lightweight frame and ergonomic handle enhance ease of movement. - Specialized materials increase manufacturing costs. - Shape-adapting handle and smart cushioning would require periodic replacement due to wear and tear. - The handle's adaptive nature requires minimal training but may need adjustment for different users. - Design does not require battery power.

Power Assisted Gurney	3.85	- Motorized wheels significantly reduce the physical effort needed for pushing and pulling. - The motors, sensors, and OBC increase costs compared to traditional gurneys but is more affordable than AGV-based solutions. - Motors require servicing but are generally reliable. - Paramedics would need training on using the throttle and power-assisted movement features. - The system is designed to last an entire EMS work shift before requiring a recharge, similar to existing powered gurneys.
Smart Suspension Gurney	2.75	- Electronically controlled dampers provide real-time terrain adaptation for smoother patient transport. - Relies on an OBC and active dampers, which consume power. - The system dynamically adjusts to terrain without requiring extensive manual control. - The additional electronic components increase the likelihood of required maintenance. - Smart suspension technology increases production and maintenance costs compared to traditional gurneys.

Appendix J: Decision Matrices and Sensitivity Analysis

Completed Decision Matrix

	Weight	AGV Assisted Gurney	Shape Adapting Handle Gurney	Power Assisted Gurney	Smart Suspension Gurney
Maneuverability	35%	4	4	5	4
Durability and Maintenance	20%	3	3	3	3
Cost	15%	2	2	3	2
Ease of Use and Training	15%	3	3	3	4
Battery Life	15%	4	4	4	3
FINAL SCORE	100%	3.35	3.35	3.85	3.35

Sensitivity Analysis Procedure

CASE	CHANGE
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Sensitivity 1	Priority was shifted from Maneuverability to Durability. The team increased the weight of Durability to 40% and distributed the difference in weight across the rest of the criterion by a value of -5%.
Sensitivity 2	Priority was shifted from Maneuverability to Cost Efficiency. The team increased the weight of Cost Efficiency to 40% and distributed the difference in weight across the rest of the criterion by a value of -6.25%.
Sensitivity 3	Priority was shifted from Maneuverability to Ease of Use and Training. The team increased the weight of Ease of Use and Training to 40% and distributed the difference in weight across the rest of the criterion by a value of -6.25%.
Sensitivity 4	Priority was shifted from Maneuverability to Battery Efficiency. The team increased the weight of Battery Efficiency to 40% and distributed the difference in weight across the rest of the criterion by a value of -6.25%.

Sensitivity Analysis 1

	Weight	AGV Assisted Gurney	Shape Adapting Handle Gurney	Power Assisted Gurney	Smart Suspension Gurney
Maneuverability	25%	4	4	5	4
Durability and Maintenance	15%	3	3	3	3
Cost	40%	2	2	3	2
Ease of Use and Training	10%	3	3	3	4
Battery Life	10%	4	4	4	3
FINAL SCORE	100%	2.95	2.95	3.6	2.95

Sensitivity Analysis 2

	Weight	AGV Assisted Gurney	Shape Adapting Handle Gurney	Power Assisted Gurney	Smart Suspension Gurney
Maneuverability	30%	4	4	5	4
Durability and Maintenance	40%	3	3	3	3
Cost	10%	2	2	3	2
Ease of Use and Training	10%	3	3	3	4
Battery Life	10%	4	4	4	3
FINAL SCORE	100%	3.3	3.3	3.7	3.3

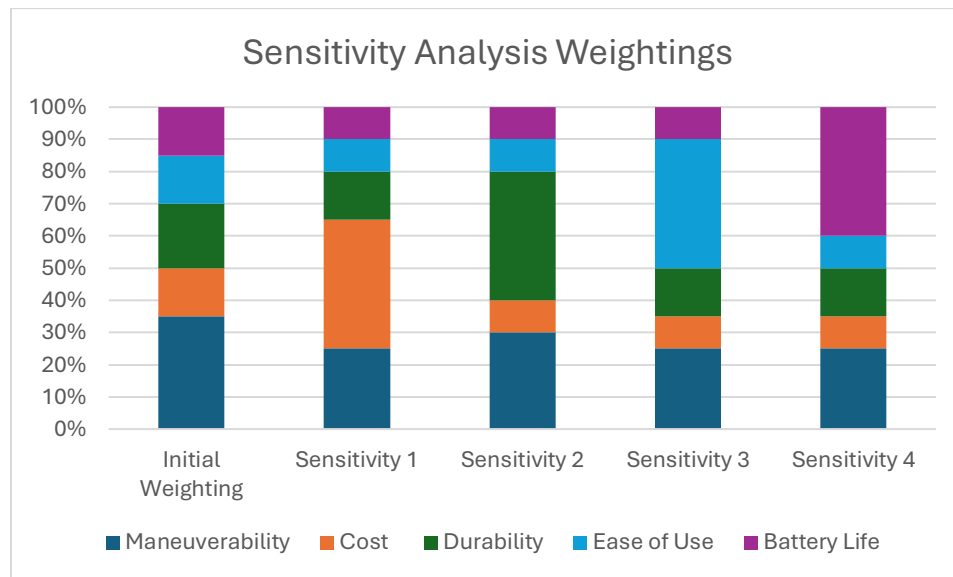
Sensitivity Analysis 3

	Weight	AGV Assisted Gurney	Shape Adapting Handle Gurney	Power Assisted Gurney	Smart Suspension Gurney
Maneuverability	25%	4	4	5	4
Durability and Maintenance	15%	3	3	3	3
Cost	10%	2	2	3	2
Ease of Use and Training	40%	3	3	3	4
Battery Life	10%	4	4	4	3
FINAL SCORE	100%	3.25	3.25	3.6	3.55

Sensitivity Analysis 4

	Weight	AGV Assisted Gurney	Shape Adapting Handle Gurney	Power Assisted Gurney	Smart Suspension Gurney
Maneuverability	25%	4	4	5	4
Durability and Maintenance	15%	3	3	3	3
Cost	10%	2	2	3	2
Ease of Use and Training	10%	3	3	3	4
Battery Life	40%	4	4	4	3
FINAL SCORE	100%	3.55	3.55	3.9	3.25

Appendix K: Sensitivity Analysis Weightings



Appendix L: Summary of Sensitivity Analysis Scores and Final Results

	AGV Assisted Gurney	Shape Adapting Handle Gurney	Power Assisted Gurney	Smart Suspension Gurney
Initial Weighting	3.35	3.35	3.85	3.35
Sensitivity 1 Score	2.95	2.95	3.6	2.95
Sensitivity 2 Score	3.3	3.3	3.7	3.3
Sensitivity 3 Score	3.25	3.25	3.6	3.55
Sensitivity 4 Score	3.55	3.55	3.9	3.25
AVERAGE SCORE	3.28	3.28	3.73	3.16
STANDARD DEVIATION	0.22	0.22	0.14	0.22

Appendix M: Derived Calculations:

Derivation of Required Torque Motor Calculation

Formula for Torque (τ)

$$\tau = r \times F_{Total}$$

Radius, r , given as 0.0762 m

Need to find Total Force (F_{Total}) exerted on the gurney

$$F_{Total} = F_{RR} + F_{Weight} - F_{Operator}$$

Need to find Rolling Resistance Force (F_{RR}), Weight Force (F_{Weight}), and Operator Force ($F_{Operator}$)

$$F_{RR} = C_{RR} \times m \times g \times \cos(\theta)$$

$$F_{RR} = 0.35 \times 157 \text{ kg} \times 9.81 \text{ m/s}^2 \times \cos(5.71)$$

$$F_{RR} = 536.38 \text{ N}$$

$$F_{Weight} = m \times g \times \sin(\theta)$$

$$F_{Weight} = 157 \text{ kg} \times 9.81 \text{ m/s}^2 \times \sin(5.71)$$

$$F_{Weight} = 153.24 \text{ N}$$

$$F_{Operator} = 194.80 \text{ N}$$

Calculate F_{Total}

$$F_{Total} = 536.38 \text{ N} + 153.24 \text{ N} - 194.80 \text{ N}$$

$$F_{Total} = 494.82 \text{ N}$$

Calculate Required Torque (τ)

$$\tau = 0.0762 \text{ m} \times 494.82 \text{ N}$$

$$\tau = 37.71 \text{ Nm}$$

Formula for Required Speed (V)

$$V = \frac{2\pi r \omega_{OUT}}{60}$$

Need to find Output Angular Speed (ω_{OUT})

Formula for ω_{OUT}

$$\omega_{OUT} = \frac{\omega_{IN}}{R}$$

Need to find Gear Ratio (R)

Formula for R

$$R = \frac{\tau_{OUT}}{\tau_{IN}}$$

In this case, τ_{OUT} is equal to the Required Torque (τ) and τ_{IN} is equal to the Stall Torque provided by the motor.

$$R = \frac{37.71 Nm}{4.69 Nm}$$

$$R = 8.1$$

Calculate (ω_{OUT})

$$\omega_{OUT} = \frac{3190 RPM}{8.1}$$

$$\omega_{OUT} = 393.83 RPM$$

Calculate V

$$V = \frac{2\pi(0.0762)(393.83)}{60}$$

$$V = 3.14 \frac{m}{s} = 11.31 \frac{km}{h} > 10 \frac{km}{h}$$

Our required Gear Ratio can provide a speed higher than the required speed.

Derivation of Battery Capacity Calculation

Calculating the required battery capacity (Ah) is essential for ensuring uninterrupted motor operation, enabling reliable patient transport. It determines how long the motors can sustain power to overcome weight and resistance, guiding battery selection. Accurate Ah calculation prevents power loss and minimizes recharging needs. This analysis accounts for peak current demands and efficiency losses. assumptions, initial conditions, engineering principles, and relevant standards are detailed in Appendix B.

Power and Current for Motors

$$P = 783.00 W, \quad V = 12.00V$$

$$P = V \times I$$

$$783.00 = 12.00 \times I$$

$$I = \frac{783.00}{12.00} = 65.25 A$$

To account for two motors in parallel, multiply the amperage by 2

$$I_{total} = 65.25 \times 2.00 = 130.5 A$$

Battery Capacity Without Optimization

$$Battery Capacity = I \times t$$

Expected to run for 1 hour:

$$Battery Capacity = 130.5 \times 1.00 = 130.5 Ah$$

Battery Capacity with Acceleration Algorithm (25% Reduction)

Assuming a 25% reduction in current due to a smooth acceleration

$$I_{opt} = 130.5 - (130.5 \times 0.25)$$

$$I_{opt} = 130.5 - 32.63 = 97.88 \text{ A}$$

Battery Capacity with optimized current:

$$\text{Battery Capacity} = 97.88 \times 1 = 97.88 \text{ Ah}$$

Battery Capacity for 60 % Full Power and 40% Half Power Usage

$$\text{Battery Capacity at Full Power } (BC_{FP}) = 97.88 \times 0.60 = 58.73 \text{ Ah}$$

$$\text{Battery Capacity at Half Power } (BC_{HP}) = 97.88 \times 0.50 \times 0.40 = 19.58 \text{ Ah}$$

Total Battery Capacity

$$\text{Total Battery Capacity } (BC_T) = BC_{FP} + BC_{HP} = 58.73 + 19.58 = 78.31 \text{ Ah}$$

Using a 60 Ah Battery Due to Weight Limitations

Due to physical limitations, 78.31 Ah is too heavy, so a 60 Ah battery is used.

Battery Capacity Equation

$$BC_T = I \times 0.60t + 0.50 \times I \times 0.40t$$

$$60.00 = 97.88 \times 0.60t + 0.50 \times 97.88 \times 0.40t$$

Calculating each term

$$60.00 = 58.73t + 19.58t$$

$$60.00 = 78.31t$$

Solving for t:

$$t = \frac{60.00}{78.31} \approx 0.76 \text{ hours}$$

$$\text{Actual Run Time} = 0.76 \text{ hours} \approx 45 \text{ minutes and 36 seconds}$$

Error Analysis

$$\begin{aligned} \text{err} &= \frac{BC_{theoretical} - BC_{actual}}{BC_{actual}} \times 100 \\ \text{err} &= \frac{78.31 - 60.00}{60.00} \times 100 \\ \text{err} &\approx 30.52\% \end{aligned}$$

Derivation of Motor Circuit Calculation

Power Formula

$$P = VI$$

Rearrange to solve for current

$$I = P/V$$

$$P = 783.00 \text{ W}, \quad V = 12.00 \text{ V}$$

$$I = \frac{783.00}{12.00} = 65.25 \text{ A}$$

$$65.25 + 65.25 = 135.25A \text{ for both motors}$$

Handle User Input in Real Time Calculation

The first calculation is done to determine the requested speed from throttle (T) based on the max speed:

$$v_{total} = T \times K_T$$

The K_T variable is set to the maximum speed of 10km/hr or 2.78 m/s conversion. When multiplying the max speed by T , the velocity requested by the user (v_{total}) is obtained. The velocity can be negative or positive depending on whether the user wants to travel forward or backwards. Next, each wheel will need to travel at different speeds during turns or rotations. The following equations focus on obtaining the necessary velocity for each of the rear wheels of the gurney:

$$v_L = v_{total} + \frac{\Omega_{total} \times d}{2}$$

$$v_R = v_{total} - \frac{\Omega_{total} \times d}{2}$$

Using the requested speed, the velocity needed for both the rear left (v_L) and right (v_R) wheels can be obtained with the differential drive equations. Value d is the width of the gurney of 0.58 m. Ω_{total} is the angular velocity needed for the wheels; this value can be negative or positive depending on the direction of turn.

The equation used to obtain the angular velocity Ω_{total} used in the previous equation is:

$$\Omega_{total} = J_x \times \Omega_{max}$$

The J_x variable represents the input from the joystick for steering the direction of the gurney. Ω_{max} can be calculated when T is 1 as the gurney will rotate on its own axis when the input is -1 or 1:

Assume v_{total} is 0 and both v_L or v_R are set to the max velocity going in opposing direction as shown in Figure 6 to solve for Ω_{total}

$$v_L = v_{total} + \frac{\Omega_{total}d}{2}$$

$$2.78 \text{ m/s} = 0 + \frac{\Omega_{total}d}{2}$$

Ω_{total} becomes Ω_{max} since the velocity of the cart 0 and it does not move from its current position

$$5.56 \text{ m/s} = \Omega_{total}d$$

$$9.59 \text{ rad/s} = \Omega_{total}$$

The next step is to determine the power needed for the motors on the wheels to maintain the requested velocity when the gurney is loaded with a patient. The rolling resistance equation and Newton's Second Law equation:

$$F_{rr} = \mu_{rr} \times m \times g$$

$$F = m \times a$$

Using another equation from Newton's Second Law and assuming motor efficiency is perfect:

$$P_{wheel} = (F + F_{rr}) \times v_L$$

The acceleration value (a) of the forward force (F) is constant. The acceleration can be found from the previously calculated required force of 494.82N, the mass load on each wheel can be assumed to be half of the total mass on each rear wheel and the acceleration is found to be:

$$494.82 \text{ N} = \frac{157 \text{ Kg}}{2} \times a$$

$$a = 6.30 \text{ m/s}^2$$

Appendix N: Pseudocode Implementation

```
BEGIN
#### INITIALIZATION ####
Initialize Serial Communication (TX/RX) With Motors
Initialize HX711 for Load Cell Data
Initialize ADC for Throttle Input
Initialize Joystick Inputs (Analog or Digital)
Initialize Motor Control Variables and Limits

#### MAIN LOOP ####
LOOP FOREVER:
    # READ SENSOR INPUTS #
    weight = ReadHX711()
    throttle = ReadADC(THROTTLE_PIN)
    joystick_x = ReadADC(JOYSTICK_X_PIN)

    # FILTER INVALID VALUES TO PREVENT UNFORESEEN BEHAVIOUR IN SYSTEM#
    throttle = DataFilter(throttle)
    joystick_x = DataFilter(joystick_x)

    # CALCULATE BASE SPEED FROM THROTTLE #
    requested_speed = throttle * MAX_SPEED

    # COMPUTE TURNING EFFECT #
    omega = joystick_x * MAX_ANGULAR_VELOCITY

    # CALCULATE MOTOR SPEEDS #
    left_motor_speed = CalculateWheelSpeed(requested_speed , omega, 'leftWheel')
    right_motor_speed = CalculateWheelSpeed(requested_speed , omega, 'rightWheel')

    # COMPUTE MOTOR POWER #
    left_motor_power = CalculateWheelPower(left_motor_speed, weight)
    right_motor_power = CalculateWheelPower(right_motor_speed, weight)
```

```

# SEND MOTOR CONTROL SIGNALS VIA SERIAL #
SerialWrite("MOTOR_LEFT:", left_motor_speed, left_motor_power)
SerialWrite("MOTOR_RIGHT:", right_motor_speed, right_motor_power)

# WAIT FOR NEXT LOOP ITERATION #
Delay(SAMPLE_RATE_MS) # Maintain stable update rate

END LOOP

### FUNCTION DEFINITIONS ###
FUNCTION ReadHX711():
    weight = GetDataFromHX711() # Read weight in kg
    RETURN weight

FUNCTION ReadADC(pin):
    value = GetAnalogInput(pin) # Read analog voltage
    RETURN value

FUNCTION DataFilter(value):
    If (value > MAX_LIMIT) # Use preset limits to filter input values outside of range
        value = MAX_LIMIT
    Else if (value < MIN_LIMIT)
        value = MIN_LIMIT
    RETURN value

FUNCTION SerialWrite(label, speed, power):
    TX_STRING = label + "," + speed + "," + power
    SendViaUART(TX_STRING)

FUNCTION CalculateWheelSpeed(speed, weight, label):
    If (label == 'leftWheel')
        motor_speed = #Apply Differential Drive Kinematics Equation for the left wheel
    Else if (label == 'rightWheel')
        motor_speed = #Apply Differential Drive Kinematics Equation for the right wheel
    Return motor_speed

FUNCTION CalculateWheelPower(speed, weight):
    Motor_power = #Apply motor power equation
    Return motor_power

```

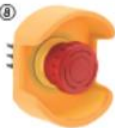
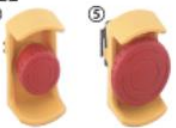
Appendix O: Response to Feedback

FEEDBACK	RESPONSE
Replace student number with student name in Signature Page	Student numbers were replaced with Student Names
Fix the incorrect categorization of constraints and criteria.	Criteria and Constraints we reassessed and double checked to ensure proper categorization.
Present constraints and Criteria as standalone sentences – No additional description between the constraint/criteria and justification.	Description between was removed and the constraints and criteria are set to standalone sentences.
Add more Criteria	Criteria were reassessed and updated as well as adding more criterion to the list.
Provide references for Cost of Labor estimates	References are added for a variety of positions and their differing labor costs.

Appendix P: Safety Features

Table: ISO 13850:2015 Compliant E-Stops

ISO13850:2015 Combination of emergency stop switch and switch guard compliant with ISO 13850:2015

Shape	Switch Guard Part No.	Applicable model	Shape	Switch Guard Part No.	Applicable model
ø16 	XA9Z-KG1	① XA1E-BV3 XA1E-BV4 ② XA1E-LV3 XA1E-LV4	ø22 	HW9Z-KG5	⑥ XW1E-BV4 XW1E-LV4 XW1E-TV4 ⑦ XW1E-BV5
ø22 		③ XW1E-BV4 XW1E-LV4 XW1E-TV4	ø30 		⑧ XN1E-BV4 XN1E-BV5 XN1E-LV4 XN1E-TV4
ø22 	HW9Z-KG4	④ XW1E-BV4 XW1E-LV4 XW1E-TV4 ⑤ XW1E-BV5			